SPEAR3 RF Station

Legacy System Interlock Architecture

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Author	Faya Wang
Reviewer	Jim Sebek and others
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Affiliation	SSRL, SLAC National Accelerator Laboratory
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This document traces every interlock path in the *legacy* SPEAR3 RF station: what trips, how fast, through which hardware, and what the software can and cannot see. It is the reference against which the upgraded protection chain in Doc 0 is justified. Where the two disagree, the upgrade document states the intended future behavior and this document states the present one. The Open Items register lists the points that still require field verification.

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Open Items — Still To Be Verified

Every finding in this document is traced to a source file or drawing. The items below are the ones that were *inferred* from that material and have not yet been confirmed against the installed hardware. Each is tagged **[O*n*]** at the point in the text where it is relied upon.

ID	Item	Required Evidence	Needed Before	Page
O1	Slot 5 MPS Shutoff module	Find the module's drawing, and record how the SPEAR MPS and orbit-interlock permits enter it and the electrical form of its J3 output, which is cabled to the Fast Interlock Chassis's spare-fault input.	Interface Chassis installation	p. 20
O2	RF MPS PLC ↔ Fast IC and panel wiring	Confirm whether any hardwired line runs from the PLC-5 to the Fast Interlock Chassis besides the AB summary and AB heartbeat that pass through the RF Local Display Panel (the detector output J31 runs the other way, to PLC-5 analog input 2 [R26]), and confirm the J3 pin order, which is reversed between SD-340-311-01 and WD-340-330-02 [R27], [R30].	Interface I/O freeze	p. 28
O3	AIM response to External MPS	The AIM's spare channel, External MPS, is enabled. Confirm on the hardware what the AIM does when it trips: whether it asserts RF_FAULT and drops the HVPS on request, as for an arc.	Interface Chassis installation	p. 21
O4	RF Local Display Panel logic	Read out the equations of the panel's GAL22V10 logic device, which forms SCR ENABLE, KLYSTRON CROWBAR and the beam abort [R29].	Interface Chassis logic design	p. 11
O5	Remote I/O word/bit map	Cross-check the assignments derived from [R9] against the deployed SLC-500 ladder program.	Controller replacement	p. 31
O6	Arc sensors	Confirm which of the ten enabled arc channels have a detector installed, and where.	Arc detection changes	p. 14

Background

This document focuses on the legacy interlock system design, providing a deeper and more detailed description of the interlock architecture within the broader SPEAR3 RF system. It is part of the overall system architecture defined in [R1] — SPEAR3 RF System Legacy Architecture, which serves as the primary system-level reference. This document extends and elaborates on the interlock-related content presented there. Relevant sections in [R1]:

- Doc L, *LLRF Controller (VXI System)* — ARC Interlock Module (AIM)
- Doc L, *HVPS Control System (B118)* — SLC-500 PLC
- Doc L, *Personnel Protection System (PPS) Interface*
- Doc L, *RF Machine Protection System (MPS)*
- Doc L, *Interlock Architecture and Signal Chain*

All source code and EPICS database files cited in this document are listed with their reference numbers in Appendix B.

How to read the latency figures in this document. The response times quoted throughout (Fast Interlock Chassis $< 1\ \mu$s, AIM interrupt service $\approx 10\ \mu$s, RF MPS PLC $\approx$ 10–15 ms, SLC-500 $\approx$ 10–20 ms, SNL state machine ≈ 1 s) are *design and analysis* values derived from the cited drawings, scan-rate settings and source code. They describe the budget of each individual layer. Consistent with [R1], no measured end-to-end installed-path latency record exists for the SPEAR3 RF interlock chain, so these figures should be treated as the expected order of magnitude rather than as commissioning data.

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1 Architecture Overview

1.1 The Five Actors

The SPEAR3 RF interlock system involves five distinct hardware/software actors. They are **not redundant layers of the same function** — each addresses a different threat class at a different speed:

Actor	Speed	Location	What It Protects Against
Fast Interlock Chassis (340-308)	< 1 μs	B132	Arc breakdown: converts the arc light into analog signals for the AIM; carries the AIM's commands to the panel
AIM Module (VXI Slot 12)	< 1 μs (HW) / ~10 μs (ISR)	VXI Crate	Compares the Fast IC's arc signals with its thresholds and latches them; watches the AB summary and heartbeat; asserts RF_FAULT on VXI backplane; forces the beam abort; captures fault waveforms; drives control outputs (HVPS_On, Filament, Solenoid) through the Fast IC to the RF Local Display Panel
RF MPS PLC (PLC-5/1771)	~10 ms	B132	Collector overpower, vacuum excursion, secondary arc, cooling
SLC-500 HVPS PLC (RIO adapter 1)	~20 ms	B118	HVPS internal faults: oil, transformer, crowbar, overvoltage, arc in HV tank
SNL State Machine (`rf_states.st,v`)	~1 s	VXI CPU (VxWorks)	Orderly shutdown sequencing, fault recording, recovery coordination

1.2 Signal Flow Diagram

Three fault sources feed five actors, as shown in Figure 1. Hardware paths (top) are independent of software; the EPICS alarm tree (bottom) is the single software trip wire to the SNL state machine.

Speed summary by path:

Path	Actor	Hardware action	EPICS notification	SNL response
Arc → Fast IC and AIM	Fast IC 340-308 and AIM	< 1 μs (RF cut by the AIM; SCR ENABLE and CROWBAR through the AIM commands and the RF Local Display Panel logic)	~50–500 μs (AIM ISR → VXI:LTCH)	~1 s (orderly HVPS shutdown)

Path	Actor	Hardware action	EPICS notification	SNL response
MPS PLC Path A → panel logic and AIM	RF MPS PLC + RF Local Display Panel + AIM	~10–15 ms once the PLC trips (HVPS kill through SCR ENABLE for collector and waveguide pressure faults; RF cut through the AB summary); the collector protection trips only after 0.70 s above its limit	~1 s (Remote I/O Path B)	~1 s (orderly HVPS shutdown, HVPS already dead)
SPEAR MPS permit / orbit interlock → Slot 5	VXI Slot 5	RF cut only; the module's response time is not documented (O1), and Slot 5 has no hardware HVPS path	~50–500 μs (AIM ISR → VXI:LTCH → STNOFF; External MPS → MPS:LTCH)	~6 s (SNL orderly shutdown); earlier only through the AIM (O3) or the collector-power cascade after 0.7 s (§5)
HVPS controller trip	SLC-500 HVPS PLC	~10–20 ms (24 V fault/enable removed from the trigger boards)	~100 ms–1 s (Remote I/O → EPICS)	~1 s (orderly RF shutdown, HVPS already off)

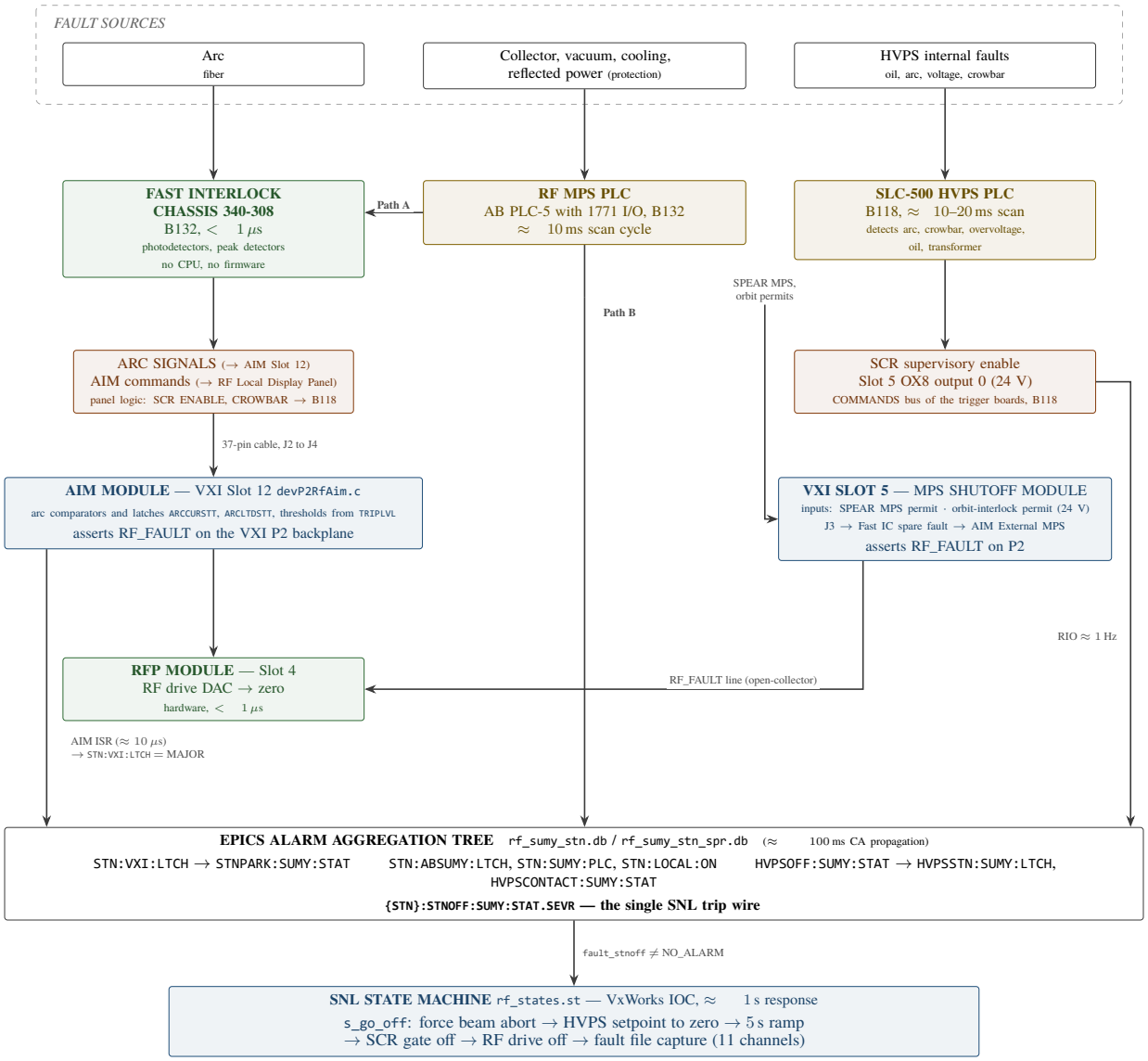


Figure 1: Legacy interlock signal flow. Three fault sources feed five actors. The hardware paths across the top are independent of software; the EPICS alarm tree at the bottom is the single software trip wire into the SNL state machine. The RF MPS PLC acts on two paths: **Path A** sends its collector and waveguide pressure summaries to the logic of the RF Local Display Panel (not drawn), which removes SCR ENABLE, and its AB summary over the panel's fibers to the Fast Interlock Chassis and the AIM; **Path B** is the Remote I/O copy of its global interlock summary, `STN:SUMY:PLC`, in the alarm tree. VXI Slot 5 takes the SPEAR MPS and orbit-interlock permits: it asserts RF_FAULT on the backplane, and its J3 output reaches the AIM through the Fast Interlock Chassis as the External MPS channel. The three HVPS fibers run between the RF Local Display Panel in B132 and the HVPS controller in B118.

2 Fast Interlock Chassis 340-308

Sources: [R1], [R6], [R19].

2.1 Function

The Fast Interlock Chassis is the analog front end of the AIM. It turns the light of the twelve arc-detector fibers into peak-detected analog signals for the AIM, whose comparators, latches and fault logic make the trip decision; it passes the AIM's six commands to the RF Local Display Panel as fibers and returns the PLC-5's AB summary and heartbeat to the AIM; and it holds the klystron forward-power detector that feeds the PLC-5. It has no processor and no power supply of its own: the AIM powers it through J15 [R32].

Physical location: B132 electronics rack, right below the VXI crate.

2.2 Inputs

Input Type	Source	Connection	Notes
Arc detection (fiber optic)	4× cavity waveguide window sensors	Fiber optic receivers on chassis	Photodetector, amplifier and peak detector per channel; the analog signal goes to the AIM's comparator [R32]
Arc detection (fiber optic)	Klystron output window sensor	Fiber optic receiver	
Arc detection (fiber optic)	Main circulator sensor	Fiber optic receiver	
RF forward power	Klystron forward coupler	Coaxial cable to the rear RF input J32; the detected level leaves on the rear output J31 (RF AMPL.) for the PLC-5	The front BNC J16 (DETECTED KLYSTRON POWER) is a monitor output and is not cabled [R26]
HVPS status (`HVPSON` in AIM `FISTAT`, bit 6)	Not documented	—	The HVPS STATUS fiber ends on the RF Local Display Panel, where it is received as the HVPS beam abort, not on this chassis [R26], [R27]; the source of the `FISTAT` bit is not documented. The AIM uses the bit to gate arc voltage readback (arc peaks not read when `HVPSON=0`).
AB summary and AB heartbeat	RF MPS PLC-5, through the RF Local Display Panel	Two fibers from the panel; passed to the AIM on J15	The AIM latches an AB summary fault and an AB heartbeat fault (Path A) [R14], [R26]
AIM commands, supplies and trip tests	AIM, cable J15	25-pin cable	The six commands, inverted and sent as fibers; the AIM's fused ± 12 V and 5 V, which power the chassis; twelve trip-test lines that drive test-light fibers [R32]

Input Type	Source	Connection	Notes
Spare fault	VXI Slot 5 MPS Shutoff module, J3 (checked November 2026)	Front LEMO	24 V input through an optocoupler that trips on a low input; passed to the AIM on the 37-pin cable as the External MPS channel (§4.3) [R32]

Signal routing: The SPEAR MPS and orbit-interlock permits are inputs of the **VXI Slot 5 MPS Shutoff module**, not of the Fast Interlock Chassis [R31], [R34]. When either permit is removed, Slot 5 asserts `RF_FAULT` on the VXI P2 backplane, cutting RF drive. Its J3 output is cabled to this chassis's spare-fault input, which passes it to the AIM as the External MPS channel. See §4.2 for the Slot 5 description.

2.3 Outputs

Output	Destination	Latency	Mechanism
AIM commands (six)	RF Local Display Panel (B132)	< 1 μs	FILAMENT_ON, SOLENOID_ON, HVPS_ON, BEAM_ABORT, FILAMENT_TIMEOUT and FAULT_RESET from the AIM, inverted and sent as fibers [R26]
AB summary, AB heartbeat	AIM Module (VXI Slot 12)	~1 μs	From the panel's fibers, on J15
Arc signals (12) and spare fault	AIM Module (VXI Slot 12)	~1 μs	Analog arc signals, differential, and the spare-fault line, on the 37-pin cable from J2 to the AIM's J4 [R32]
Detected klystron power	RF MPS PLC-5, analog input 2, channel 10 (terminal 12)	—	Rear output J31 (RF AMPL.); the front BNC J16 is not cabled [R26]
Test light (12)	Arc-detector fibers	—	Fiber transmitters J33 to J44, driven by the AIM's trip-test lines [R32]

SCR ENABLE and KLYSTRON CROWBAR are not outputs of this chassis. They are formed by a programmable logic device (GAL22V10) in the RF Local Display Panel 340-310 from the HVPS on request and the VXI beam abort (the AIM's commands, through this chassis), the HVPS STATUS fiber, the main and bucking solenoid faults and the PLC-5's collector and waveguide pressure summaries. They leave the panel on fibers to the trigger interconnect boards of the HVPS controller in B118; STATUS returns from there to the panel. The same logic drives the beam abort [R26], [R27], [R29], [R30]. The equations of the logic are not documented. **[O4]**

These latencies are signal-propagation times, not fault-clearing times. The distinction matters when reasoning about fault energy:

Stage	Time
Fiber command changes (AIM, Fast IC, panel logic)	μ**s scale** (not measured)
Crowbar thyristor begins conducting	$\approx$ **10** μ**s** after the trigger
Primary AC current interrupted	**4–8 ms**
Vacuum contactor opens (PPS Chain 1)	$\approx$ **200 ms** worst case

Removing SCR gate drive does **not** stop conduction immediately — the thyristors continue to the next current zero. The crowbar is what bounds energy into the klystron: **< 5 J with it, < 40 J without** (SLAC-PUB-7591).

2.4 Front Panel

The front panel of the Fast Interlock Chassis (SLAC drawing 340-308; front and rear photographed in Doc L) shows:

- Lamps for the six AIM commands: **FILAMENT ON**, **SOLENOID ON**, **HVPS ON**, **BEAM ABORT**, **FILAMENT TIMEOUT** and **FAULT RESET**; each is lit while its fiber to the RF Local Display Panel is lit, so **BEAM ABORT** is lit when there is no abort
- **TEST CH.** 1 to 12 lamps for the AIM's arc trip tests
- **INPUT MON.** CH.1 through CH.12 monitor points of the arc channels
- **SPARE FAULT** input (LEMO), cabled to J3 of the VXI Slot 5 MPS Shutoff module
- **J16 DETECTED KLYSTRON POWER** — BNC monitor output of the RF detector, not cabled; the RF input J32 and the output J31 to the PLC-5 are on the rear panel [R26]

2.5 Trip Mechanism

Each arc channel has a threshold that the AIM sets in software. When a fiber-optic arc pulse arrives (arc sensors generate an optical burst on breakdown):

1. The Fast IC's photodetector and peak detector pass the pulse to the AIM as an analog signal
2. The AIM's comparator for that channel fires against its threshold, set by a DAC from the `TRIPLVL` register [R32]
3. The AIM latches the arc (`ARCLTDSTT`, held until a fault reset) and asserts `RF_FAULT`
4. The AIM's six command signals are set by its register bits and by its on-board fault logic [R31]; when they drop the HVPS on request or force the beam abort, the RF Local Display Panel logic removes SCR ENABLE and, as its equations specify, fires the crowbar (O4)

Source: [R1] (interlock architecture and signal chain); [R21]; [R32].

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3 AIM Module (VXI Slot 12)

Sources: [R3], [R6], [R7], [R14], [R21].

3.1 Role Clarification — AIM vs. Fast Interlock Chassis

The AIM (schematic SD-340-309-01) and the Fast Interlock Chassis 340-308 (schematics SD-340-308-01 and -02) are companion units designed as a pair. A spare Fast Interlock Chassis of a slightly different version, 340-307, sits above the RF Local Display Panel in rack B132-12 and is not in use; the unit in service is the 340-308.

Fast Interlock Chassis 340-308 is the **analog front-end**:

- Directly receives fiber-optic arc sensor pulses from cavity windows, klystron output, and circulator
- Converts the arc light into analog signals, one photodetector, amplifier and peak detector per channel; the threshold comparison is made on the AIM
- Passes the AIM's six command signals to the RF Local Display Panel as fibers and returns the PLC-5's AB summary and heartbeat to the AIM; SCR ENABLE and KLYSTRON CROWBAR are formed by the panel (§2.3)
- Sends the twelve analog arc signals and the spare-fault line to the AIM over a 37-pin cable from its J2 to the AIM's J4 [R32]

The AIM (schematic SD-340-309-01) is the **VXI digital companion**:

- Receives the twelve analog arc signals and the spare-fault line on its J4, and compares each arc signal with its own threshold [R32]
- Its 12-channel arc registers (`ARCCURSTT`, `ARCLTDSTT`) hold the states of its own comparators
- **Sets the comparator thresholds**: writing to the AIM's `TRIPLVL` registers (A24 address space) sets the DAC that drives the comparator of each channel on the AIM [R32]. Enforced range: 10–255 (MINTRIPLVL / MAXTRIPLVL in `devP2RfAim.c`).
- Asserts `RF_FAULT` on the VXI P2 backplane (sub-μs) to cut RF drive from the RFP module
- Generates the ISR $\rightarrow$ EPICS alarm chain notification (~10 μs)
- Drives six control signals, which the Fast Interlock Chassis converts to fibers for the RF Local Display Panel: station online request, filament and focus request, HVPS on request, VXI beam abort, filament timeout bypass and reset [R26], [R31]
- Forces the beam abort and resets it; the panel's beam-abort output is not cabled at SPEAR3 (§3.8)
- **Captures arc-channel history waveforms** via on-module 512 KB ADC ring buffer (12-ch AIM `HISBUF`): the AIM's fast ADC continuously samples all arc channel voltages and the HVPS voltage into this buffer; on fault the buffer freezes and is read back
- Captures RF/IQA fault waveforms at the SNL level (11 channels $\rightarrow$ `/dat/FAULT*_N` files)

Key distinction: The AIM makes the arc decision — it compares each channel's analog signal with its threshold, latches per-channel arc events and asserts the VXI-level RF protection. It does **not** receive the fiber-optic pulses directly: the Fast IC 340-308 converts and peak-detects them [R32].

3.2 Hardware Identity

- **Module type**: SLAC PEP-II RF custom VXI module — "Arc detector / Interlock Module" (AIM), schematic SD-340-309-01
- **VXI slot**: 12 (last slot, rightmost in crate)
- **Device support driver**: `devP2RfAim.c` (1,982 lines)
- **Primary EPICS record**: `{STN}:STN:AIM:MODU` (type `p2RfAim`)
- **DAS instruction files**: `/tbl/aimDas0.inst`, `/tbl/aimDas1.inst`
- **History file**: `/dat/aimHist.dat`

3.3 Arc Detection (12 Channels)

The AIM module has 12 independently configurable arc detection channels. Each channel has:

- **Fault enable**, set by a jumper on the AIM (headers H5 and H6; no jumper = enabled) and read back as `ARCENBSTT` (`MODU.AFES`); EPICS cannot change it [R3], [R32]
- **Threshold** setting (8-bit value, range 10–255): configured via EPICS writing to the AIM's `TRIPLVL` registers (A24 offset `0x0020` for ch 0, successive offsets per channel). The device driver (`devP2RfAim.c`) writes the value to the AIM VXI register, which sets the DAC of that channel's comparator on the AIM [R32].
- **Current status** (`ARCCURSTT`, `MODU.ACST`) — real-time state of the AIM's comparator: 1 = clear, 0 = tripped
- **Latched status** (`ARCLTDSTT`, `MODU.ALST`) — 1 = clear, 0 = tripped since the last fault reset. The driver masks both status words with the enables, so a disabled channel reads 0 [R3]

The SPEAR3 database names eleven channels [R35]. The AIM's fault-enable word read `0x71FB` (29179) on 11 November 2026, which enables exactly these eleven:

Bit	Channel (latch record)	Fault enable, 11 November 2026
0	Klystron arc (`KLYS:ARC:LTCH`)	Enabled
1	Circulator arc (`CIRC:ARC:LTCH`)	Enabled
2	Not named	Disabled
3–6	Cavity 1–4 window arc, air side (`CAV`*n*`AIR:ARC:LTCH`)	Enabled
7, 12–14	Cavity 1–4 window arc, vacuum side (`CAV`*n*`VACM:ARC:LTCH`)	Enabled
8	External MPS, the spare input (`STN:MPS:LTCH`; §4.3)	Enabled
9–11	Not channels	—
15	Not named	Disabled

Note on channel count: Ten arc channels are enabled, at six locations: the klystron, the circulator and the four cavity windows, each window on its air and vacuum sides. The enables are jumpers, so they do not show whether a detector is connected; which channels have one has not been confirmed by inspection. **[O6]**

3.4 Hardware Trip Path — RF_FAULT Backplane Line

This is the fastest path, entirely in hardware with no software involvement (Figure 2):

The `RF_FAULT` line is **open-collector**: any module on the VXI P2 bus can assert it. The RFP module's reaction is purely analog — no DSP, no SNL, no EPICS involved.

3.5 ISR Path — Software Notification (~10 μs)

After the hardware trip fires:

```
AimIsr() {
    read FAST INTERLOCK STATUS REGISTER  // latched, not ISR register (Sass 2004 fix)
    update rec->istt
    board->intPrc++
    scanOnce(rec)   // schedules AIM:MODU for EPICS processing
}
```

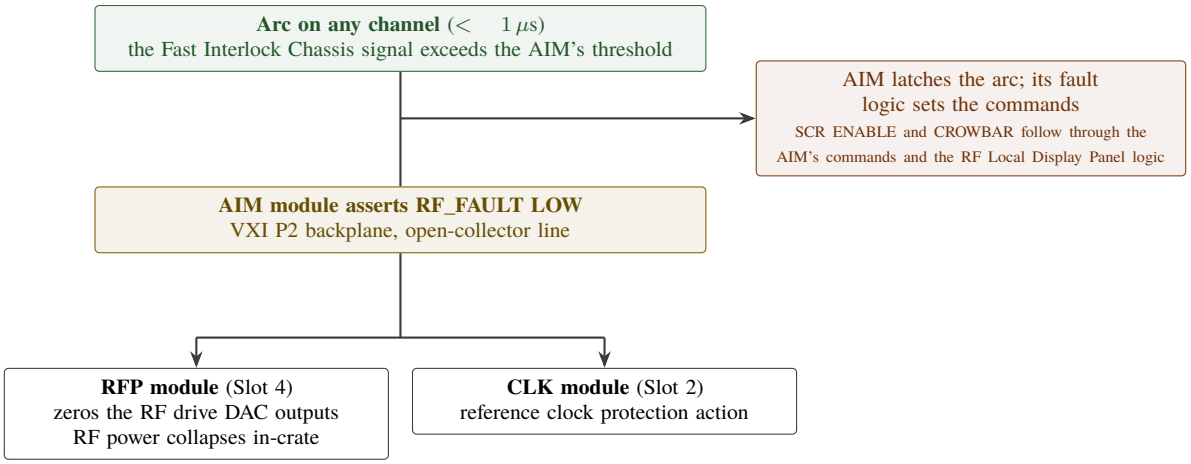


Figure 2: Arc trip path. The Fast Interlock Chassis converts the arc light; the AIM's comparators and latches decide and assert RF_FAULT. No processor is in the path.

Critical 2004 fix (Sass, `rf_interlock_vxi.db` rev 1.2): Prior to January 2004, the VXI interlock records read the AIM **interrupt status register** (ISR). Because EPICS processed these records continuously, the ISR bits self-cleared on every read — a latched fault could be invisible to the alarm system because it was cleared before alarm propagation. The fix changed to reading the AIM **fast interlock status/latch register**, which holds bits until a fault reset (`MODU.RSTF`) clears them. This was a significant reliability fix.

The record `{STN}:STN:AIM:MODU` then processes via a three-stage fanout (`FANO1` → `FANO2` → `FANO3`) that updates all derivative records.

3.6 AIM Status PV Map

PV	AIM Register	Content
`{STN}:STN:AIM:ARCCURSTT`	`MODU.ACST`	mbbiDirect — arc current status, one bit per channel (1 = clear)
`{STN}:STN:AIM:ARCLTDSTT`	`MODU.ALST`	mbbiDirect — arc latched status, one bit per channel (1 = clear, 0 = latched until a fault reset)
`{STN}:STN:AIM:ARCENBSTT`	`MODU.AFES`	mbbiDirect — fault-enable word, set by jumpers on the AIM; `0x71FB` on 11 November 2026
`{STN}:STN:AIM:FSTFLT`	`MODU.FFLT`	mbbiDirect — first-fault arc channel
`{STN}:STN:AIM:FSTINTSTT`	`MODU.FSTT`	mbbiDirect — fast interlock status register: VXI backplane, AB heartbeat, AB summary and forced faults, latched; HVPS on; spare fault [R6], [R14]
`{STN}:STN:AIM:INTSTATE`	`MODU.ISTT`	mbbiDirect — overall AIM interlock state
`{STN}:STN:AIM:INTACK`	`MODU.IACK`	mbboDirect — interlock acknowledge

These feed into `rf_interlock_vxi.db`:

```
STN:STN:AIM:LTCH.BN  →  STN:STN:VXI:LTCH  (bi, MAJOR alarm)
    → STN:STNPARK:SUMY:STAT → STN:STNOFF:SUMY:STAT   (trip)
    → STN:STNMPS:SUMY:LTCH.INPC                       (status)
```

3.7 AIM Control Output Signals (Fiber Optic Outputs)

The AIM module drives **6 control output signals** via the `FICTRL` (Fast Interlock Control) register (A16 address `0x20`). They are set by the SNL state machine and by the AIM's on-board fault logic [R31]; the filament and solenoid commands are set by the operator. They leave the AIM on J3, pass through the Fast Interlock Chassis (J15), which inverts them and converts them to light, and reach the RF Local Display Panel as six fibers [R26]:

Signal Name	EPICS PV	AIM FICTRL bit	Destination	Function
Klystron Filament On	`{STN}:STN:AIM:FILAMENT`	bit 2 (`MODU.FOO`)	RF Local Display Panel (filament and focus request), then the PLC-5	Starts the PLC-5's filament warm-up and enables the focus supplies; required before HVPS startup
Solenoid On	`{STN}:STN:AIM:SOLENOID`	bit 0 (`MODU.SOO`)	RF Local Display Panel (station online request), then the PLC-5	With the online key switch, sets the PLC-5's station-online bit (`STN:LOCAL:ON`); required before HVPS startup
HVPS On (ai-mon)	`{STN}:STN:AIM:MODU.HVPS`	bit 1 (`MODU.HVPS`)	RF Local Display Panel logic (HVPS on request)	**Hardware enable gate** — AIM holds LOW at startup; SCR ENABLE cannot be lit until IOC asserts it in `HVPSONSUB`
Force Beam Abort (fba)	`{STN}:STN:AIM:FRCBMABT`	bit 6 (`MODU.FBA`)	RF Local Display Panel (VXI beam abort); its beam-abort output is not cabled	Station not ready for beam; asserted in `s_go_off` Step 1
Fault Reset	`{STN}:STN:AIM:MODU.RSTF`	bit 5 (`MODU.RSTF`)	AIM latches; RF Local Display Panel and PLC-5 reset	Clears the AIM's arc and fast-interlock latches; through the fiber, resets the panel's latches and the PLC-5's
Filament Timeout	`{STN}:STN:AIM:MODU.FTOR`	bit 3 (`MODU.FTOR`)	RF Local Display Panel (filament timeout bypass), then the PLC-5 as a 24 V request	Filament timer bypass; not commonly used in normal SPEAR3 operation

Note: `Forced_Fault` (bit 4, `MODU.FF`, PV `{STN}:STN:AIM:FRCFLT`) is the AIM's own forced fault, set from EPICS, not one of the six fibers, and `Beam_Abort_Reset` (bit 7) resets the AIM's beam-abort latch without a fiber of its own [R26], [R31].

Strobed signals (HVPS, FILTMOOVRD, FLTRESET, RSTBMABT) are written high then immediately re-cleared in the same processing cycle to avoid latching; latching signals (SOLENOID, FILAMENT, FRCDFLT, FRCBMABT) hold their state.

3.8 Beam Abort

The state machine forces a beam abort (`fba`) whenever it turns the station off or enters TUNE (`TUNESUB`), and resets it when the station reaches ON or PARK with no faults [R2]. The forced abort leaves the AIM as the VXI beam abort fiber, through the Fast Interlock Chassis, to the RF Local Display Panel. The panel's logic combines it with the HVPS STATUS fiber (the HVPS beam abort) and the main and bucking solenoid faults, latches the result until reset and drives a rear-panel BNC output (J6) [R26], [R29]. The output was made for the PEP-II beam-abort system [R33]; at SPEAR3 J6 is not cabled (checked November 2026), and the SPEAR MPS takes no status from the RF in hardware: it dumps the beam by removing the RF permit through VXI Slot 5. The beam abort therefore acts only inside the station: on the AIM's latch (`STN:BEAMABORT:STAT`), the panel's lamps and the ripple loop, which sets no new DC coefficient until the abort is reset. The code's comments call this reset "resetting BATS".

BATS reset sequence (required before restarting):

```
/* RESET_BMABTSUB macro in rf_states.st */
if (fault_stnoff == NO_ALARM) {
    fba = 0;        pvPut(fba)   // de-assert Force Beam Abort
    pvPut(rba)                   // Reset Beam Abort (RBA → MODU.RBA)
    pvPut(rstf)                  // Reset Faults (RSTF → MODU.RSTF --- clears arc latches)
}
```

This three-write sequence is only executed when `fault_stnoff == NO_ALARM` — all fault conditions must be clear first.

3.9 Fault Capture — Two Independent Systems

There are **two completely separate** capture systems that activate on fault. They capture different signals by different mechanisms at different speeds.

3.9.1 System A — AIM Hardware History Buffer (12-channel AIM only)

The 12-channel AIM module includes a 512 KB on-module ring buffer (`HISBUF`, A24 address `0x0038`, `AIM_K_HISBUFID`). An **on-board fast ADC of the AIM** continuously samples all arc channel voltages and the HVPS voltage into this memory [R32]. Two `FICTRL` bits control the ADC:

FICTRL bit	Name	0	1
bit 8	`ADCMUX`	HVPS voltage + all 12 arc channels (normal)	HVPS voltage only
bit 9	`ADCCTL`	Start / continue buffering (normal; no fault)	Stop buffering (freeze on fault)

Normal operation: `ADCCTL=0` and no fault → ADC continuously overwrites the ring buffer.
On fault: The fault assertion sets `ADCCTL=1` → buffer freezes at fault instant → pre-fault arc waveforms and HVPS voltage are preserved.

Buffer readout: The driver reads arc voltage peak values via the `ARCVOL` register (A24 `0x003A`) — a sequential-readout register that returns one channel's voltage per read, cycling through all channels. Similarly, `CRTVOL` (A24 `0x003C`) returns per-channel crate voltage measurements. The readout is gated: if `FISTAT.HVPSON = 0` (HVPS off), the driver skips arc voltage readback (no HV → no expected arc peaks).

7-channel AIM (legacy PEP-II): Used a different mechanism — two DAS (Data Acquisition System) chips with instruction files (`/tbl/aimDas0.inst`, `/tbl/aimDas1.inst`) that defined what registers to capture. This DAS mechanism is NOT used in the 12-channel SPEAR3 AIM. The `/dat/aimHist.dat` file listed in §3.2 is the 7-ch DAS artifact; SPEAR3 uses the HISBUF hardware buffer.

3.9.2 System B — SNL Fault File Capture (`ss rf_statesFF`)

A software-level capture triggered by `efSet(ffwrite_ef)` in the SNL `s_go_off` state at ~1 s after the hardware fault. Captures **11 RF/IQA signal channels** (not arc voltages). The `ss rf_statesFF` state set:

1. Increments fault slot number (circular buffer 1–15, `NUMFAULTS=15`)
2. Timestamps via `pvTimeStamp(hvpswdefault)` — the HVPS voltage PV's timestamp
3. Places 11 signal modules in LOAD mode
4. Reads hardware signal memory buffers from RFP, IQA modules
5. Writes to `/dat/FAULTxxx_N` files (N = fault slot 1–15)
6. Maximum wait: `MAXFFWAIT = 180` × `20 ticks` ≈ `3.6 s`

The 11 fault file channels:

Channel	File Pattern	Signal
0	`FAULTRfpSI_N`	RFP Signal I (cavity forward, I)
1	`FAULTRfpSQ_N`	RFP Signal Q (cavity forward, Q)
2	`FAULTRfpCI_N`	RFP Cavity I (cavity voltage, I)
3	`FAULTRfpCQ_N`	RFP Cavity Q (cavity voltage, Q)
4	`FAULTIqa1Amp_N`	IQA1 amplitude waveform
5	`FAULTIqa2Amp_N`	IQA2 amplitude waveform
6	`FAULTGvf_N`	GVF signal (PEP-II heritage, inactive in SPEAR3)
7	`FAULTAim_N`	AIM arc/interlock status snapshot
8	`FAULTCmbI_N` / `FAULTCmbQ_N`	Combiner I/Q (PEP-II, inactive in SPEAR3)
9	`FAULTIqa3Amp_N`	IQA3 amplitude waveform
10	*(additional)*	Additional channel added in 2003 Laznovsky revision

Source: [R2] lines 1900–2200 (`ss rf_statesFF`); [R3]; [R21]; [R6] (`AIM_A_HISBUF`, `AIM_V_ADCMUX`, `AIM_V_ADCCTL`).

3.10 Inputs

Input	Source	Connection	Notes
Arc signals (12 channels)	Fast Interlock Chassis 340-308	37-pin cable, Fast IC J2 to AIM J4	Analog, peak-detected; compared on the AIM with its thresholds; `ARCCURSTT` and `ARCLTDSTT` hold the comparator states, the latter until a fault reset [R32]

Input	Source	Connection	Notes
Spare fault (External MPS)	VXI Slot 5 J3, through the Fast IC's spare-fault input	37-pin cable, pin 34	Bit 8 of the arc registers and of the fast interlock status register (`FSTINTSTT`, `MODU.FSTT`); `STN:MPS:LTCH` (§4.3) [R6], [R32], [R35]
AIM interrupt	AIM on-board IRQ logic	VXI interrupt line	`AIM_M_ARCDETD` (arc detected), `AIM_M_VXIBPFLT` (VXI backplane fault), `AIM_M_FRCDFLTINT` (forced fault), etc.
AB summary and AB heartbeat	RF MPS PLC-5, through the RF Local Display Panel fibers and the Fast IC	J15 of the Fast IC to J3 of the AIM [R26]	The AIM latches an AB summary fault and, when the pulses stop, an AB heartbeat fault, in its fast interlock status register [R14]
AIM_M_SOFTWARE interrupt	EPICS IOC write	VXI register write	Software-triggered interrupt (for test/forced-fault operations)

3.11 Outputs

Output	Destination	Mechanism	Notes
`RF_FAULT` assertion (open-collector LOW)	VXI P2 backplane	VXI bus	Sub-μs; causes RFP (slot 4) to zero RF drive DAC and CLK (slot 2) to react
HVPS_On fiber (`aimon`, `MODU.HVPS`)	RF Local Display Panel logic (HVPS on request $\rightarrow$ SCR ENABLE)	Through the Fast IC, then fiber	Software gate; must be explicitly asserted by SNL `HVPSONSUB`; AIM holds LOW at startup
Klystron Filament On (`MODU.FOO`)	RF Local Display Panel (filament and focus request), then the PLC-5	Through the Fast IC, then fiber	Set by the operator, not by the SNL; the filament must be warm before HVPS turn-on
Solenoid On (`MODU.SOO`)	RF Local Display Panel (station online request), then the PLC-5	Through the Fast IC, then fiber	Set by the operator, not by the SNL; the station-online bit it sets is required before HVPS can be energized
Force Beam Abort (`MODU.FBA`, `fba`)	RF Local Display Panel (VXI beam abort); its beam-abort output is not cabled	Through the Fast IC, then fiber	Station not ready for beam; asserted in `s_go_off` Step 1
Reset Beam Abort (`MODU.RBA`, `rba`)	AIM beam-abort latch	AIM register write	De-asserts the forced beam abort; used in recovery `RESET_BMABTSUB`
Fault Reset (`MODU.RSTF`, `rstf`)	AIM hardware latches	AIM register write	Clears `ARCLTDSTT`, `AIMARC:LTCH` and the fast interlock latches

Output	Destination	Mechanism	Notes
EPICS PVs (arc status, interlock state)	EPICS IOC (VxWorks)	EPICS CA record processing	`ARCCURSTT`, `ARCLTDSTT`, `ARCENBSTT`, `FSTFLT`, `INTSTATE` → `STN:VXI:LTCH` → alarm tree

4 VXI Slot 5: MPS Shutoff Module

Sources: [R5], [R17], [R18], [R19], [R20], [R31], [R34], [R35].

4.1 What It Is

This slot requires careful distinction between hardware labeling, software configuration, and functional role.

The SRF1 crate configuration names slot 5 **"MPS Shutoff"** and slot 6 "Link Passthru"; the crate template `crat_vxi_13slot.template,v` takes the slot names as macros [R17], [R35]. The module in slot 5 has a connector J3 that is cabled to the Fast Interlock Chassis (checked November 2026). J. Dusatko describes it as the MPS interlock module: it receives the interlock signal of the SPEAR MPS and drives `RF_FAULT` on the VXI backplane; it has no VXIbus interface and takes only crate power and `RF_FAULT` from the backplane [R31]. No drawing of the module is in the repository. **[O1]**

The IOC software (`rf_vxi_modules_All.substitutions,v`) [R18] loads `cf2.db` (the Comb Filter 2 database) for slot 5 — **this is a PEP-II inherited template artifact**. The `devP2RfCf2.c` driver (2,970 lines) is a PEP-II-only module; the CF2 DSP functionality (comb filter, group delay equalizer, filter bank) is **not exercised in SPEAR3**.

Aspect	PEP-II	SPEAR3 SRF1
Physical slot 5 module	CF2 (Comb Filter 2) DSP	MPS Shutoff module
Software DB loaded	`cf2.db`	`cf2.db` (PEP-II template artifact)
Device driver active?	Yes — `devP2RfCf2.c`	No — physical CF2 module not installed
Functional use	RF comb filter feedback	MPS permit interface

Source: [R19] line 65; [R20] line 95.

4.2 Hardware Function

The MPS Shutoff module in slot 5 is the station's input for the SPEAR3 machine permits:

1. It uses the VXI P2 backplane only for power and `RF_FAULT` [R31]
2. Its inputs are the two machine permits: the SPEAR MPS and the orbit interlock each give the LLRF a 24 V permit [R34]. Their wiring into the module is not on any RF drawing in the repository (O1). The RF MPS PLC does not feed it: the PLC-5's output drawings show no line to the VXI crate [R36]
3. When a permit is lost, the module asserts `RF_FAULT`: the RFP removes the drive at once, and the AIM latches the backplane fault (`STN:VXI:LTCH`), which turns the station off through the state machine [R14], [R31]

4. Its J3 output is cabled to the Fast Interlock Chassis's SPARE FAULT input, which passes it to the AIM's J4 on pin 34 of the 37-pin cable [R32]; the AIM holds it as its spare channel, External MPS (§4.3)

Key routing fact: The SPEAR MPS and orbit-interlock permits do **not** reach the RF Local Display Panel, and they reach the Fast Interlock Chassis only as the module's J3 output on its spare-fault input. Their removal cuts RF drive at once through `RF_FAULT`, but Slot 5 has no path that removes SCR ENABLE or fires the crowbar. A hardware HVPS kill can come only from the AIM, if its fault logic acts on the External MPS channel as on an arc (O3), or from the collector-power cascade (§5). Otherwise the `RF_FAULT` assertion propagates through the EPICS alarm chain (`STN:VXI:LTCH` MAJOR → `STNPARK:SUMY:STAT` MAJOR → `STNOFF:SUMY:STAT` MAJOR → `fault_stnoff` in the SNL state machine), which drives `s_go_off` and performs an orderly HVPS shutdown approximately 6 seconds later.

4.3 Software Visibility

The module has no VXIbus interface, so no EPICS record reads it [R31]. A Slot 5 trip appears in EPICS in two ways:

- `{STN}:STN:VXI:LTCH`, the AIM's latch of the backplane fault in its fast interlock status register [R14]. This is the record that trips the station, through `STNPARK:SUMY:STAT` and `STNOFF:SUMY:STAT` [R11]. It does not tell which module asserted `RF_FAULT`.
- `{STN}:STN:MPS:LTCH`, "External MPS": the AIM's spare channel (arc-register bit 8), which carries the J3 output. SPEAR3 took over this channel, reserved for MPS, in 2003 [R35]. The record shows the AIM's latch of the channel when the channel's fault enable is set, and reads OK otherwise. It feeds `STNMPS:SUMY:LTCH` [R12], a station status summary; it is not an input of `STNPARK`, `STNON` or `STNOFF`, and the state machine does not read it [R2], [R11]. The AIM's first-fault register has no bit for it [R6].

The channel is enabled: the AIM's fault-enable word `STN:AIM:ARCENBSTT` read `0x71FB` on 11 November 2026, with bit 8 set (§3.3). The enables are jumpers on the AIM board. What the AIM's fault logic does when this channel trips is not documented. **[O3]**

The CF2 OVFL and status records (`{STN}:STN:CF2OVFL:STAT`) loaded for slot 5 are not connected to any active hardware in SPEAR3 (their `DTYP="Soft Channel"` reads from the inactive `{STN}:STN:CF2:MODU` record).

4.4 Key Conclusion

VXI Slot 5 in SPEAR3 is the station's input for the SPEAR MPS and orbit-interlock permits. It does NOT run the CF2 DSP firmware; the cf2.db PV records loaded for slot 5 are inactive (hardware not present). Removal of either permit causes the RFP module to cut RF drive at hardware speed through `RF_FAULT` and — via the EPICS alarm chain (`STN:VXI:LTCH` → `STNOFF:SUMY:STAT` → `fault_stnoff`) — the SNL state machine performs an orderly HVPS shutdown ~6 s later. The module's J3 output reaches the AIM as the External MPS channel, which records the cause. Slot 5 itself produces no **immediate hardware** HVPS shutdown (no SCR ENABLE removal, no crowbar); the AIM may, if its fault logic acts on the External MPS channel (O3). With the drive cut and the HVPS on, the klystron collector also takes the full cathode power, and the RF MPS PLC removes SCR ENABLE through the RF Local Display Panel logic if that power stays above its 1 200 kW limit for 0.70 s (§5).

4.5 Analysis: Is the MPS Shutoff Module Necessary?

The MPS PLC acts in hardware through the RF Local Display Panel: its collector and waveguide pressure summaries feed the panel logic, and its AB summary reaches the AIM over the panel and Fast IC fibers. The SPEAR MPS and orbit-interlock permits enter Slot 5, independently of the RF MPS PLC. When the RF MPS PLC trips, or when either permit is removed, the following consequences apply:

Path	Triggered By	Immediate HVPS Action	Delayed HVPS Action (SNL)	RF Drive Action
Path A — RF Local Display Panel and AIM (§6.5)	Collector or waveguide pressure summary drops; AB summary fault	**SCR ENABLE removed by the panel logic** (fiber → HVPS controller, B118, ~10–15 ms); crowbar as the logic specifies (O4)	Orderly ramp-to-zero + `hvpstrig=OFF` (~1–2 s, but HVPS already dead)	`RF_FAULT` via AIM → VXI backplane
Slot 5 — SPEAR MPS or orbit permit	Either permit removed	**None** from Slot 5 — it has no connection to the SCR ENABLE or crowbar fibers; the AIM may act on its External MPS channel (O3)	Orderly HVPS shutdown via SNL `s_go_off` (~6 s: `VXI:LTCH` → `STNOFF` → `fault_stnoff`)	`RF_FAULT` asserted on VXI P2 backplane

Assessment:

For the SPEAR MPS and orbit-interlock permits, Slot 5 is the **exclusive** RF kill path; the Fast Interlock Chassis only passes its J3 output to the AIM. Their removal cuts RF drive immediately (via VXI backplane) and also triggers an orderly HVPS shutdown via the SNL state machine approximately 6 seconds later through the following alarm chain:

```
Slot 5 RF_FAULT → AIM ISR (~10 μs) → STN:VXI:LTCH (MAJOR)
    → STNPARK:SUMY:STAT (MAJOR) → STNOFF:SUMY:STAT (MAJOR)
    → fault_stnoff ≠ NO_ALARM → s_go_off (SNL rf_states.st)
        → HVPS voltage setpoint → 0 (ramp, 5 s)
        → hvpstrig = OFF  (STN:HVPSSCR:ON:CTRL = 0)
```

The key distinction between fault sources is **how quickly** the HVPS is shut down:

Fault source	Immediate HVPS hardware kill from this path	HVPS shutdown time
Arc (Fast IC and AIM)	Through the AIM's commands and the panel logic, as their logic specifies (O4)	μs scale if so; otherwise through the cascade or the SNL
RF MPS PLC (Path A)	Yes for collector and waveguide pressure faults — SCR ENABLE removed by the panel logic once the PLC trips; other faults through the cascade	~10 ms after the PLC trips; the collector protection itself waits 0.70 s
SPEAR MPS / orbit (Slot 5)	**No** from Slot 5 itself; possibly from the AIM's External MPS channel (O3)	~6 s (SNL orderly shutdown), or 0.7 s through the collector-power cascade if the collector power stays above its limit (§5)

Critically, Slot 5 does **not** independently produce an immediate HVPS shutdown. The hardware HVPS kill (SCR ENABLE removal, and the crowbar if the logic so specifies) comes from the RF Local Display Panel logic responding to its own inputs: the AIM's commands, if its fault logic acts on the External MPS channel (O3), or the RF MPS PLC's collector summary. With drive = 0 and HVPS on, the klystron collector

absorbs full cathode power; the PLC latches collector overpower once that power has stayed above 1 200 kW for 0.70 s and drops its collector summary (Path A), and the panel logic then removes SCR ENABLE. See §5 for the full mechanics and signal chain.

Why Slot 5 is essential — not merely retained:

Slot 5 is **architecturally necessary** for SPEAR3 RF protection:

1. **Sole integration point for SPEAR MPS and orbit interlock**: These two permits are wired only to Slot 5. Without Slot 5, their removal would have zero effect on either RF drive or HVPS — the klystron would continue transmitting into the cavity regardless of the machine beam-permit status or orbit feedback state. No other wiring path or RIO register carries them; EPICS sees them only through the AIM, as `STN:VXI:LTCH` and the External MPS channel.
2. **Provides the only VXI-backplane RF kill for machine-level permits**: The RF Local Display Panel and the SLC-500 HVPS PLC have no knowledge of the SPEAR MPS or orbit-interlock permits, and the Fast IC only passes Slot 5's J3 output to the AIM. Only Slot 5 bridges these machine-level signals into the RF station protection architecture.
3. **Carried forward from PEP-II architecture**: The CF2 slot served a different function in PEP-II; the physical slot was repurposed in SPEAR3 for the machine permits and is integral to SPEAR3 machine protection integration.

Conclusion: VXI Slot 5 is **essential** to the SPEAR3 RF protection architecture. It is the **exclusive** integration point for the SPEAR MPS and orbit-interlock permits into RF station protection. Removal of either permit results in: (1) immediate hardware RF drive cut via VXI backplane, (2) the External MPS indication in the AIM, and (3) orderly HVPS shutdown via SNL `s_go_off` ~6 s later. Slot 5 itself has no connection to the SCR ENABLE or crowbar fibers; an earlier hardware HVPS kill can come only from the AIM (O3) or from the collector-power cascade after 0.7 s (§5).

---

5 Cascade Physics: Secondary Trips Following RF Drive Cutoff

Sources: [R13], [R16], [R23], [R37].

When RF drive is cut by a SPEAR MPS permit withdrawal or orbit interlock trip (via Slot 5, or by any mechanism that sets the RFP drive DAC to zero), the immediate RF cutoff creates a secondary physical condition that can fire the RF MPS PLC's collector overpower protection. The PLC-5 trips on it only after the collector power has stayed above 1 200 kW for 0.70 s [R37], so this cascade removes SCR ENABLE about 0.7 s after the RF cut, if the high voltage has not been lowered below the limit first. That is earlier than the SNL removes the SCR supervisory enable (~6 s), but it is not a fast path.

5.1 Mechanism 1 — Klystron Collector Overpower (High Certainty)

A klystron is a velocity-modulation amplifier: it converts cathode beam power into RF output power. When RF drive drops to zero, the klystron immediately stops amplifying — the bunched electron beam produces no RF output. However, as long as the HVPS remains energized, the cathode beam current continues to flow through the klystron body and is deposited in the collector. All cathode input power therefore becomes collector heat dissipation:

$$P_{\text{collector}} = P_{\text{cathode}} - P_{\text{RF out}} \approx P_{\text{cathode}} \quad (\text{when RF drive} = 0,\ \text{HVPS on})$$

During normal CW operation, the klystron operates at some efficiency $\eta = P_{\text{RF out}}/P_{\text{cathode}}$, so the collector absorbs only the fraction $(1-\eta)$ of cathode power. When drive is cut, $P_{\text{RF out}} \to 0$ and the collector

must suddenly absorb 100% of cathode power — a step increase proportional to the operating efficiency. For a klystron running at 50% efficiency, this is a 2× step in collector power dissipation.

The RF MPS PLC monitors collector power continuously via a dedicated AB analog input channel:

EPICS PV	Record	Source	Description
`{STN}:KLYSCOLLPLC:POWER`	`ai`	AB Remote I/O T4[41], EGUF=1200 kW	Real-time PLC collector power reading (HIHI=1175 kW MAJOR, HIGH=1150 kW MINOR)
`{STN}:KLYSCOLL:POWER:ULIM`	`ai`	AB Remote I/O T4[42], EGUF=1200 kW	The PLC's trip limit, read back: fixed at 1 200 kW in the program, with a 0.70 s delay [R37]
`{STN}:KLYSCOLL:POWER`	`sub` (subIQpowerNet)	INPA=`HVPS:POWER`, INPB=`KLYSOUTFRWD:POWER`	EPICS-computed collector power = cathode power − klystron RF output
`{STN}:KLYSCOLL:POWER:LTCH`	`bi`	AB Remote I/O WL=16, WF=31, B=2	Hardware latch bit set by PLC when it trips on collector overpower

Important distinction: `KLYSCOLLPLC:POWER.HIHI = 1175 kW` is an **EPICS software alarm** that propagates through Channel Access; it does not directly change the PLC outputs. The PLC computes the collector power itself, as the HVPS voltage times the current minus the square of 0.017 times the RF detector reading, and latches collector overpower when that stays above 1 200 kW for 0.70 s; the latch drops the collector summary to the RF Local Display Panel and the PLC's global interlock summary, independently of EPICS [R37].

Cascade signal chain (Figure 3):

5.2 Mechanism 2 — Reflected Power Transient (Conditional)

When klystron output power drops abruptly, the sudden impedance change at the output waveguide may produce a brief transient in reflected power at the klystron output junction. Operational calibration data records `{STN}:KLYSOUTREFL:POWER` (IQA1 Ch2) HIHI = 10.8 W, HIGH = 8.1 W — these are the EPICS IQA-based software alarm thresholds.

No hardware comparator watches reflected power in the legacy station: the Fast Interlock Chassis has no reflected-power inputs [R32], so a transient acts only through the IQA software thresholds above.

However, this mechanism is conditional and less certain than Mechanism 1:

- **Circulator isolation**: The circulator between the klystron output and the cavities is specifically designed to route reflected power (including cavity ring-down) to a load port, providing ~20 dB or better isolation to the klystron. A well-functioning circulator absorbs the transient.
- **Software only**: the IQA thresholds act through EPICS, at the IQA's update rate, not in hardware.
- **Drive-level dependence**: If the klystron was operating at reduced power before the trip, the reflected power transient may be too small to exceed the IQA threshold.

Conclusion: Mechanism 1 (collector overpower) is the dominant secondary trip: it fires 0.70 s after an RF drive cutoff when the HVPS stays energized at normal operating power, where the cathode power (about 1.4 MW at 72 kV and 19.4 A [R1]) exceeds the 1 200 kW limit. Mechanism 2 (reflected power transient) is a possible additional software trigger but is not guaranteed.

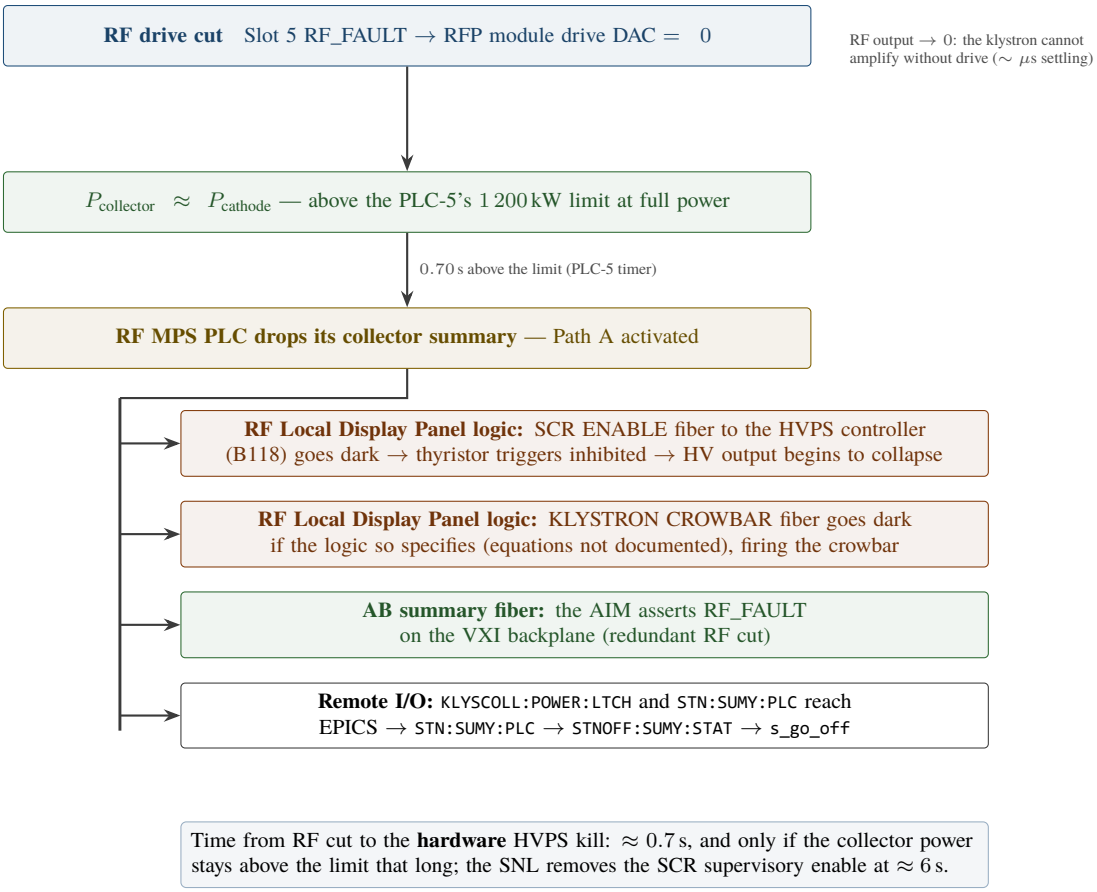


Figure 3: Cascade following an RF drive cut. Secondary trips follow the primary event and can reach the HVPS faster than the SNL shutdown does.

5.3 Net Effect on HVPS Kill Timing

Scenario	HVPS kill mechanism	Time from SPEAR MPS / orbit trip
Software path only (no cascade)	SNL `s_go_off` Step 4: `hvpstrig=OFF`	~6 s
Via the AIM's External MPS channel, if its fault logic drops the HVPS on request	RF Local Display Panel logic: SCR ENABLE removed	μs scale (not documented, O3)
Via collector overpower cascade (normal operation)	RF MPS PLC collector summary → RF Local Display Panel logic: SCR ENABLE removed (crowbar as the logic specifies)	**~0.7** s, if the collector power stays above 1 200 kW
Via reflected power transient (conditional)	No hardware path: only the IQA software alarm	Software time scale (if it fires)

Fault record implication: The cascade trigger appears in the fault record as `{STN}:KLYSCOLL:POWER:LTCH` (collector overpower), not as `STN:MPS:LTCH` or `STN:VXI:LTCH`. The root cause (SPEAR MPS or orbit trip) is visible in those latter PVs, about 0.7 s earlier. At the default ~1 Hz archiver resolution the two may fall in the same or adjacent samples — event-driven archiver timestamps are required for precise sequencing.

6 RF MPS PLC (PLC-5/1771)

Sources: [R13], [R23].

6.1 Platform

Attribute	Detail
Hardware	Allen-Bradley PLC-5 with 1771 I/O (upgraded from PLC-5 / 1771-DCM)
Location	B132 (same rack as VXI crate)
Remote I/O adapter	**Adapter 3** (`STNDCM`)
Status	In service. A ControlLogix replacement is assembled and was tested without RF power; it is not in service [R1]

6.2 Inputs

Input	Source	Connection	Notes
Klystron collector power (calculated)	Klystron high-voltage and RF output detectors	Analog I/O module in the PLC-5 1771 rack	Cathode power minus RF output; excessive difference = klystron overheating
Reflected power (4 cavities)	RF detector diodes at cavity and waveguide junctions	Analog I/O	VSWR or arc condition in waveguide
Secondary arc sensors	Arc sensor inputs not in Fast IC path	Digital I/O	Redundant arc coverage in waveguide runs
Cooling water flow	Flow meters on klystron and dummy load circuits	Digital/analog I/O	Below minimum flow trips MPS
Cooling water temperature	Temperature sensors on water circuits	Analog I/O	Exceeding maximum trips MPS
Klystron vacuum	Ion pump current or Penning gauge	Analog I/O	Vacuum excursion protection
HVPS fault status	SLC-500 HVPS PLC	Remote I/O (the VXI scanner polls both adapters independently)	HVPS faults forwarded to MPS summary
476 MHz reference power	`{STN}:STNREF:POWER:LTCH`	EPICS gateway	Reference RF monitor
RF Local Display Panel key switches	RF Local Display Panel 340-310 (J2)	24 V lines through cross-connect X530	Local/remote and online/offline modes [R29], [R30]

6.3 Outputs

Output	Destination	Connection	Notes
Collector and waveguide pressure summaries (Path A)	RF Local Display Panel logic (J3)	24 V lines through cross-connect X530	Dropping either removes SCR ENABLE (and fires the crowbar as the panel logic specifies, O4) [R27], [R29]
AB summary and AB heartbeat (Path A)	AIM, through the RF Local Display Panel fibers and the Fast IC	24 V lines (J3), then fiber	AB summary fault → AIM → `RF_FAULT` [R26]
15 area summaries	RF Local Display Panel display (J3)	24 V lines through X530	Status indication only
Remote I/O data table	EPICS IOC (via AB-6008 scanner)	RIO words `T6[W,B]`	Software path (Path B) — the global interlock summary `{STN}:STN:SUMY:PLC` (table 6, word 35, bit 0), an input of `STNOFF:SUMY:STAT`, and the fault latches [R9], [R11]

6.4 Protection Functions

The MPS PLC monitors **equipment protection** parameters on a ~10 ms scan cycle:

Monitored Parameter	Trip Condition	Notes
Klystron collector power	Cathode power − RF output exceeds limit	Excessive collector power = poor klystron efficiency → overheating
Reflected power	Reflected power at any cavity exceeds limit	VSWR/arc condition in waveguide
Arc conditions	Secondary arc sensors (not in Fast IC path)	Redundant arc coverage
Cooling water flow	Below minimum flow rate	Klystron and waveguide loads
Cooling water temperature	Exceeds maximum	Thermal runaway detection
Klystron vacuum	Ion pump current or Penning gauge exceeds limit	Vacuum excursion protection
HVPS fault conditions	Status from SLC-500 via Remote I/O	Forwarded to MPS summary
Reference power level	`{STN}:STNREF:POWER:LTCH`	476 MHz reference monitor

6.5 Trip Actions — Two Paths

Path A — Hardware HVPS shutdown + RF cut (fast, ~10 ms PLC scan):

The MPS PLC's hardwired outputs run as 24 V lines through cross-connect X530 to the RF Local Display Panel [R26], [R30]; whether any other hardwired line reaches the Fast Interlock Chassis has not been traced. [O2] When the PLC detects a fault requiring immediate action:

1. The PLC drops the affected summaries and its AB summary
2. The collector and waveguide pressure summaries are inputs of the RF Local Display Panel logic, which removes SCR ENABLE on the fiber to the HVPS controller in B118 and, as its equations specify, fires the crowbar [R27], [R29]
3. The AB summary reaches the AIM over the panel's fiber and the Fast Interlock Chassis → AIM asserts `RF_FAULT` on VXI backplane → RFP RF drive zeroed

This path: ~10 ms (PLC scan) + panel logic and fiber response (μs scale) ≈ **~10–15 ms total** once the PLC has decided; the collector protection decides only after 0.70 s above its limit (§5). **Action**: HVPS shutdown (SCR ENABLE) for the faults wired into the panel logic, AND RF drive cut. Other MPS faults, for example cooling or vacuum [R27], remove high voltage only through the collector-power cascade (§5) or the SNL.

Path B — Software/EPICS notification (slow, ~1 s):

1. The PLC's global interlock summary and fault latches change in its Remote I/O data table
2. VXI AB6008 scanner reads Remote I/O registers at ~1 Hz
3. Updates `{STN}:STN:SUMY:PLC` and the latch records [R9]
4. `STN:SUMY:PLC` is an input of `{STN}:STNOFF:SUMY:STAT.SEVR` [R11]
5. SNL `fault_stnoff != NO_ALARM` triggers `s_go_off`

Path B is **bookkeeping** — the MPS hardware has already acted on Path A. Path B ensures the EPICS state machine learns about the fault, logs it, triggers fault file capture, and performs the orderly shutdown sequence. **Action**: Orderly HVPS ramp-to-zero + fault file capture (hardware already tripped via Path A).

Note on SPEAR MPS and orbit interlock: These two permits enter VXI Slot 5, independently of the RF MPS PLC. Their removal triggers `RF_FAULT` on the VXI backplane but does **not** trigger Path A. The `RF_FAULT` line asserted by Slot 5 is seen by the AIM module's ISR, which propagates through the EPICS alarm chain (`STN:VXI:LTCH` → `STNPARK:SUMY:STAT` → `STNOFF:SUMY:STAT` → `fault_stnoff`) and causes the SNL `s_go_off` to execute an orderly HVPS shutdown ~6 s later; Slot 5's J3 output also sets the AIM's External MPS channel. The RF MPS PLC does not relay these signals. See §4.2 for the Slot 5 inputs and §4.5 for the alarm chain analysis.

Note: The immediate HVPS shutdown is owned by Path A through the RF Local Display Panel logic; Path B only informs the state machine.

6.6 EPICS PV Path

Figure 4 shows how a PLC trip reaches the state machine through Remote I/O (Path B).

6.7 MPS Wiring Documents

33 MPS wiring diagrams (SLAC drawings wd3403300200 through wd3403303400) describe the complete MPS hardware signal chain. Original harness is for the PLC-5.

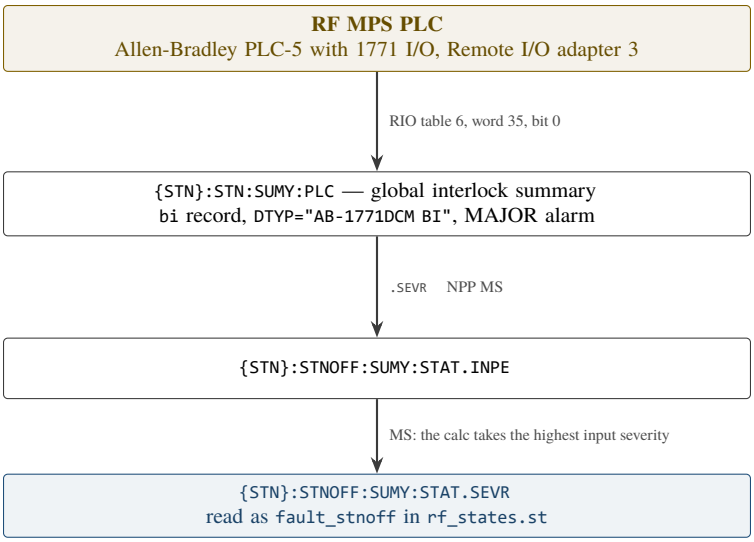


Figure 4: RF MPS PLC (Allen-Bradley PLC-5 with 1771 I/O): its global interlock summary reaches the SNL trip wire through Remote I/O (Path B).

7 SLC-500 HVPS PLC

Sources: [R4], [R8], [R9], [R10], [R15], [R16], [R23], [R24], [R25].

7.1 Platform

Attribute	Detail
CPU	Allen-Bradley 1747-L532
Remote I/O adapter module	Allen-Bradley 1747-DCM, chassis Slot 1, full rack
Location	B118 (HVPS building), Hoffman Box enclosure
Remote I/O adapter	**Adapter 1** (`HVPSDCM`, full rack)
Rack	10-slot SLC chassis (slots 0–9; slot 4 empty)

7.2 Inputs

Input	Source	Connection	Notes
HVPS DC output voltage	HVPS power supply analog	Slot 8–9 Analog I/O (AI word 30)	Used for voltage readback and overvoltage detection
HVPS DC output current	HVPS power supply analog	Slot 8–9 Analog I/O (AI word 31)	Used for current readback and overcurrent detection
Primary AC current	AC current transformer	Slot 8–9 Analog I/O	Overcurrent detection (`HVPSAC:CURR:LTCH`)
Oil temperature (thermocouple)	Oil tank thermocouple	Slot 3 thermocouple input	`HVPSOIL:TEMP:LTCH` overtemperature latch

Input	Source	Connection	Notes
PPS enable	PPS system (GOB1208PNE, Pin E→F)	Slot 6 IB16 digital input 14	Used for HV vacuum contactor control
SCR ENABLE and KLYSTRON CROWBAR fibers	**RF Local Display Panel (B132)**	**HFBR-2412 receivers on the right-side and left-side trigger interconnect boards**	**The boards act on the fibers directly: a dark SCR ENABLE inhibits the thyristor triggers and a dark KLYSTRON CROWBAR fires the crowbar [R27], [R28]. The boards' monitoring bus carries their state (SCR ENABLE MON and others); its PLC input assignment is part of O5. The HVPS STATUS fiber is an output of the left-side board (§7.3).**
Crowbar fired / arc status	Internal HVPS self-monitoring	Slot 6 IB16 digital inputs	`HVPS:CROWBAR:LTCH`, `HVPSKLYS:ARC:LTCH`, etc.
Oil level	Float switch in HV tank	Slot 6 IB16 digital input	`HVPSOIL:LEVEL:LTCH`
Transformer arc	Arc detector inside HV transformer	Slot 6 IB16 digital input	`HVPSXFORM:ARC:LTCH`
SCR bank status	SCR gate driver feedback	Slot 6 IB16 digital inputs	`HVPSSCR1:ON:STAT`, `HVPSSCR2:ON:STAT`
Contactor status	HV vacuum contactor auxiliary contacts	**Slot 7 IV16** digital inputs 0–3	Blocking relay (IN0), overcurrent relay (IN1), contactor closed (IN2), contactor ready (IN3) — `HVPSCONTACT:*:STAT` group. These sit on the **IV16 in slot 7**, not on the slot-6 IB16
`hvpstrig` (SCR enable command)	EPICS IOC via Remote I/O (SNL `HVPSONSUB`)	Remote I/O incoming data table → Slot 5 OX8 output 0	Supervisory SCR enable gate; must be ON for HVPS to produce HV

7.3 Outputs

Output	Destination	Connection	Notes
24 V fault/enable (`HVPSSCR:ON:CTRL`)	Trigger interconnect boards (COMMANDS bus), same B118 enclosure	Slot 5 OX8 output 0; no fiber [R28]	Supervisory thyristor-trigger enable; ~20 ms speed; must be on together with the SCR ENABLE fiber from the RF Local Display Panel
HVPS STATUS fiber	RF Local Display Panel (B132), received as the HVPS beam abort	HFBR-1412 transmitter on the left-side trigger interconnect board	Lit while the 24 V fault/enable is high and no crowbar is commanded; it says nothing about whether the thyristors fire [R27], [R28]
Ross grounding switch command	Ross Engineering HQ3 grounding switch	Slot 2 IO8 (120 VAC)	**PPS Chain 2**: grounds HV tank before personnel entry
HV vacuum contactor command	Ross Engineering HQ3 vacuum contactor	Slot 5 OX8 OUT2 → relay chain (K4 → MX → RR → L1)	**PPS Chain 1**: connects/disconnects 12.47 kV AC to HVPS primary (fail-safe open)

Output	Destination	Connection	Notes
HVPS Remote I/O status registers	EPICS IOC (via AB-6008 scanner, adapter 1)	Remote I/O	All `HVPS*:*:LTCH` and `HVPS*:*:STAT` PVs populated from these registers
Voltage setpoint output	HVPS voltage control SCR firing angle	Slot 8–9 Analog I/O (AO)	SNL `rf_hvps_loop.st,v` writes ramp-controlled setpoint

7.4 Rack Slot Configuration

Slot	Module	Function
0	1747-L532 CPU	Program execution
1	1747-DCM	Remote I/O adapter module — presents the SLC-500 chassis to the VXI scanner as adapter 1 (full rack)
2	IO8 (8-pt, 120 VAC)	Ross grounding switch output (PPS Chain 2)
3	Thermocouple input	HVPS temperature measurements
5	OX8 relay output	Output 0: 24 V fault/enable to the trigger boards (see §7.6); output 2: contactor relay chain
6	IB16 digital input	PPS enables, HVPS status inputs
8–9	Analog I/O	HVPS voltage readback, current readback, setpoint output

7.5 What the SLC-500 Interlocks On

The SLC-500 monitors every significant HVPS internal condition and exposes them as EPICS records via Remote I/O. The full set of HVPS interlock sources, with their Remote I/O word/bit assignments derived from `rf_digital_All.substitutions,v` [R9] — these have not been cross-checked against the deployed ladder program [O5]:

7.5.1 Category 1 — Electrical Faults

PV	RIO Card/Bit	Alarm	Description
`{STN}:HVPS:CROWBAR:LTCH`	C12.9	MAJOR	**Crowbar fired** — HVPS crowbar thyristor discharged; stored energy dumped
`{STN}:HVPS:VOLT:LTCH`	C14.1	MAJOR	**Overvoltage** — DC output exceeded rated voltage
`{STN}:HVPSAC:CURR:LTCH`	C12.3	MAJOR	**AC current trip** — primary AC current exceeded limit
`{STN}:HVPS:OPENLOAD:LTCH`	C14.9	MAJOR	**Open load** — no current path to load (klystron disconnected or contactor open during energization)

PV	RIO Card/Bit	Alarm	Description
{STN}:HVPS:PANIC:LTCH	C12.11	MAJOR	**Emergency Off (PANIC)** — manual or automatic emergency shutdown latch
{STN}:RFHVPS:CROWBAR:LTCH	C14.11	MAJOR	**RF crowbar request** — request from RF system to fire HVPS crowbar

7.5.2 Category 2 — Oil System

PV	RIO Card/Bit	Alarm	Description
{STN}:HVPSOIL:LEVEL:LTCH	C12.14	MAJOR	**Oil level low** — insulating oil in HV tank below minimum
{STN}:HVPSOIL:TEMP:LTCH	C12.15	MAJOR	**Oil temperature high** — insulating oil overtemperature
{STN}:HVPSOIL:TEMP	Analog (SLC-500 AI)	Monitored	Oil temperature value (°C); also has configurable ULIM at {STN}:HVPSOIL:TEMP:ULIM

Design note: HVPS oil temperature was removed from the HVPSSTN:SUMY:LTCH roll-up in the Sass 2004 revision (rf_sumy_hvps.db,v rev 1.2 log: "Remove HVPS oil temperature from HVPS summary. It remains in the overall temperature summary"). Oil temperature LTCH still contributes to {STN}:HVPSTEMP:SUMY:LTCH → {STN}:STNTEMP:SUMY:LTCH path.

7.5.3 Category 3 — Transformer / HV Tank

PV	RIO Card/Bit	Alarm	Description
{STN}:HVPSXFORM:ARC:LTCH	C14.10	MAJOR	**Transformer arc** — arc detector inside the HV transformer tank
{STN}:HVPSXFORM:PRESS:LTCH	C14.6	MAJOR	**Transformer overpressure** — SF or oil pressure in transformer enclosure exceeded limit
{STN}:HVPSXFORM:VACM:LTCH	C14.7	MAJOR	**Transformer vacuum/pressure** — vacuum integrity or gas pressure fault
{STN}:HVPSKLYS:ARC:LTCH	C12.4	MAJOR	**Klystron arc** (as seen from HVPS side) — arc current surge into cathode load
{STN}:HVPS:TEMP:LTCH	C14.0	MAJOR	**Overtemperature** — general HVPS enclosure over-temperature

7.5.4 Category 4 — Supply and Contactor Status

PV	RIO Card/Bit	Alarm	Description
{STN}:HVPS:PPS:STAT	C14.8	MAJOR on 0=NOPERMIT	**PPS Status** — PPS permit present (PERMIT=0, NOPERMIT=1 → alarm inverted)
{STN}:HVPS12KV:VOLT:STAT	C12.1	MAJOR on 0=NOTAVAIL	**12 kV AC available** — primary 12.47 kV supply status
{STN}:HVPSAC:POWER:STAT	C12.2	MAJOR on 0=OFF	**Auxiliary AC power** status
{STN}:HVPSDC:POWER:STAT	C12.10	MAJOR on 0=OFF	**Auxiliary DC power** status
{STN}:HVPSSUPPLY:ON:STAT	C14.4	MAJOR on 0=OFF	**HV supply on** status
{STN}:HVPSSUPPLY:READY:STAT	C14.5	MAJOR on 0=NOTREADY	**HV supply ready** status
{STN}:HVPSENERFAST:ON:STAT	C12.12	MAJOR on 0=OFF	**Fast inhibit** — energization fast inhibit active
{STN}:HVPSENERSLOW:START:STAT	C12.13	MAJOR on 1=INACTIVE	**Slow start** — energization slow-start sequencer
{STN}:HVPSCONTACT:CLOSE:STAT	C12.5	MAJOR on 0=OPEN	HV contactor close status
{STN}:HVPSCONTACT:ON:STAT	C12.6	MAJOR on 0=OFF	HV contactor on status
{STN}:HVPSCONTACT:OPEN:STAT	C12.7	MAJOR on 1=OPEN	HV contactor open status
{STN}:HVPSCONTACT:READY:STAT	C12.8	MAJOR on 0=NOTREADY	HV contactor ready
{STN}:HVPSSCR1:ON:STAT	C14.2	MAJOR on 0=OFF	SCR bank 1 on status
{STN}:HVPSSCR2:ON:STAT	C14.3	MAJOR on 0=OFF	SCR bank 2 on status

Note on Remote I/O addressing: ”C12.9” means Card (output word) 12, bit 9. Word 12 = physical I/O module card 12÷2 = slot 6. The SLC-500 uses 8-bit word addressing where adjacent card pairs share a 16-bit word.

7.5.5 Category 5 — Analog Monitoring

PV	Source	Alarm Limits	Description
{STN}:HVPS:VOLT	SLC-500 AI	HIHI=87, HIGH=85 kV	HVPS DC output voltage
{STN}:HVPS:CURR	SLC-500 AI	HIHI=30, HIGH=30 A	HVPS DC output current

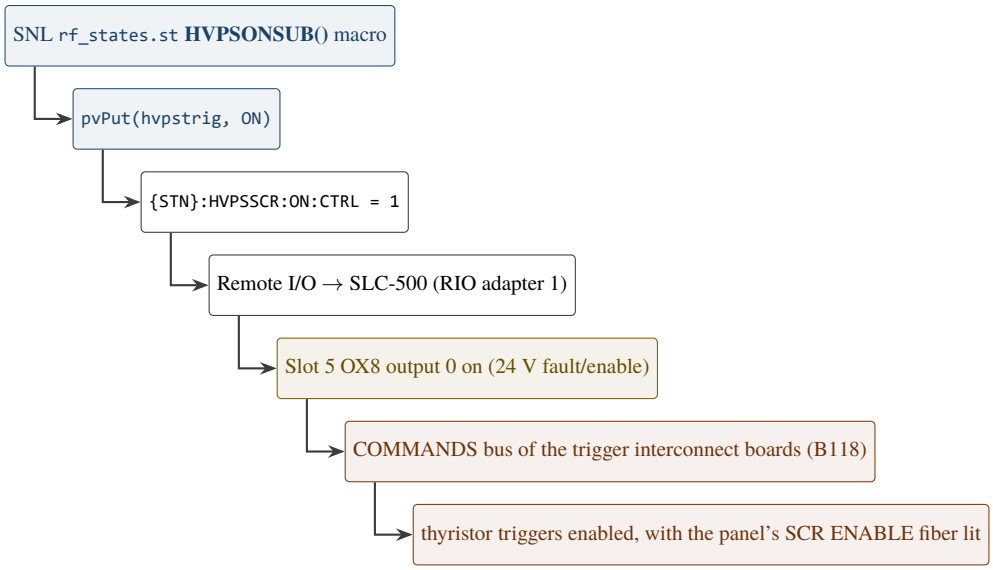


Figure 5: HVPS enable chain executed by the `HVPSONSUB()` macro in `rf_states.st`.

PV	Source	Alarm Limits	Description
`{STN}:HVPS:PLC:VOLT`	SLC-500 AI word 30	±100 kV range	Voltage readback from PLC (independent ADC)
`{STN}:HVPS:PLC:CURR`	SLC-500 AI word 31	±50 A range	Current readback from PLC (independent ADC)
`{STN}:HVPSAC:CURR`	SLC-500 AI	Monitored	Primary AC current
`{STN}:HVPSOIL:TEMP`	Thermocouple slot 3	ULIM configurable	Oil temperature (°C)

7.6 SCR Enable Output — The HVPS Supervisory Enable Path

The SLC-500 enables the thyristor triggers through output 0 of its slot-5 relay module (24 V fault/enable), which drives the COMMANDS bus of the trigger interconnect boards in the same B118 enclosure; no fiber is involved [R28]. This is the **supervisory SCR ENABLE path** — it must be held on for the HVPS to maintain output voltage, and the HVPS STATUS fiber is lit only while it is on. Figure 5 shows the sequence that establishes it.

This path is **distinct from** the SCR ENABLE fiber formed by the RF Local Display Panel:

Path	Speed	Source	Notes
RF Local Display Panel → SCR ENABLE (fiber)	μs	Panel logic	Hardware permit, removed on fault
SLC-500 output 0 → COMMANDS bus (24 V)	~20 ms	Supervisory enable/disable	Must be explicitly enabled; cleared in s_go_off

Both act on the trigger interconnect boards in B118: the panel's path is a fiber from B132, the SLC-500's is wired on the COMMANDS bus. **Both** must be asserted for the thyristors to fire. The panel path is the

fast-trip path; the SLC-500 path is the supervisory gate that must be deliberately enabled by the SNL state machine as part of the startup sequence.

PV: `{STN}:HVPSSCR:ON:CTRL` (bo record, `rf_hvps.db,v`): ”HVPS SCR Enable Switch”. Writing ON enables the HVPS; writing OFF disables (part of s_go_off orderly shutdown).

7.7 HVPS Alarm Aggregation Tree

Figure 6 shows how the individual HVPS latches roll up into the three summary records.

7.8 PPS Dual-Chain Architecture (Contactor and Grounding Switch)

	Chain 1 — HV Vacuum Contactor	**Chain 2 — Ross Grounding Switch**
Safety function	Disconnect 12.47 kV AC primary power from HVPS	Ground the HVPS HV bus (−77 kV) for safe personnel access
SLC-500 output	Slot-5 OX8 OUT2 relay contact → K4 → MX → L1 relay chain	Slot-2 IO8 OUT3 (120 VAC) → Ross switch coil
PLC rung	Rung 0017 (mislabeled ”Crowbar On” on original drawing — corrected to ”Contactor Enable”)	Rung 0016 (Ross switch enable)
PPS input	PPS 1 (GOB1208PNE Pin E→F)	PPS 2 (GOB1208PNE Pin G→H)
Operating state	Coil held energized (contactor **CLOSED** — 12.47 kV connected)	Coil held energized (switch held **OPEN** — HV bus not grounded)
Fail-safe state	PPS removed → K4 drops → L1 drops → contactor **spring-opens** → 12.47 kV disconnected	PPS removed → 120 VAC removed → switch **spring-closes** → HV bus grounded
Read-back signal	S5 NC auxiliary contact → GOB1208PNE Pins A-B	Ross switch NC auxiliary contact → GOB1208PNE Pins C-D
Read-back meaning	Pins A-B shorted = contactor OPEN (12.47 kV removed, safe)	Pins C-D shorted = switch CLOSED (HV bus grounded, safe)

7.8.1 Chain 1 — HV Vacuum Contactor Relay Path

The full relay chain from PPS to contactor, as routed through the SLC-500. This is the personnel-safety path described in `tex/docL-legacy-architecture/L_legacy_system_architecture.pdf` (PPS interface).

```
PPS Enable (GOB1208PNE, Pin E→F)
    → SLC-500 Slot-6 IB16 Input 14
    → PLC Rung 0017 (ladder logic)
    → Slot-5 OX8 OUT2
    → Terminal Strip TS-5 → switchgear cable
    → K4 relay (PPS control)
```

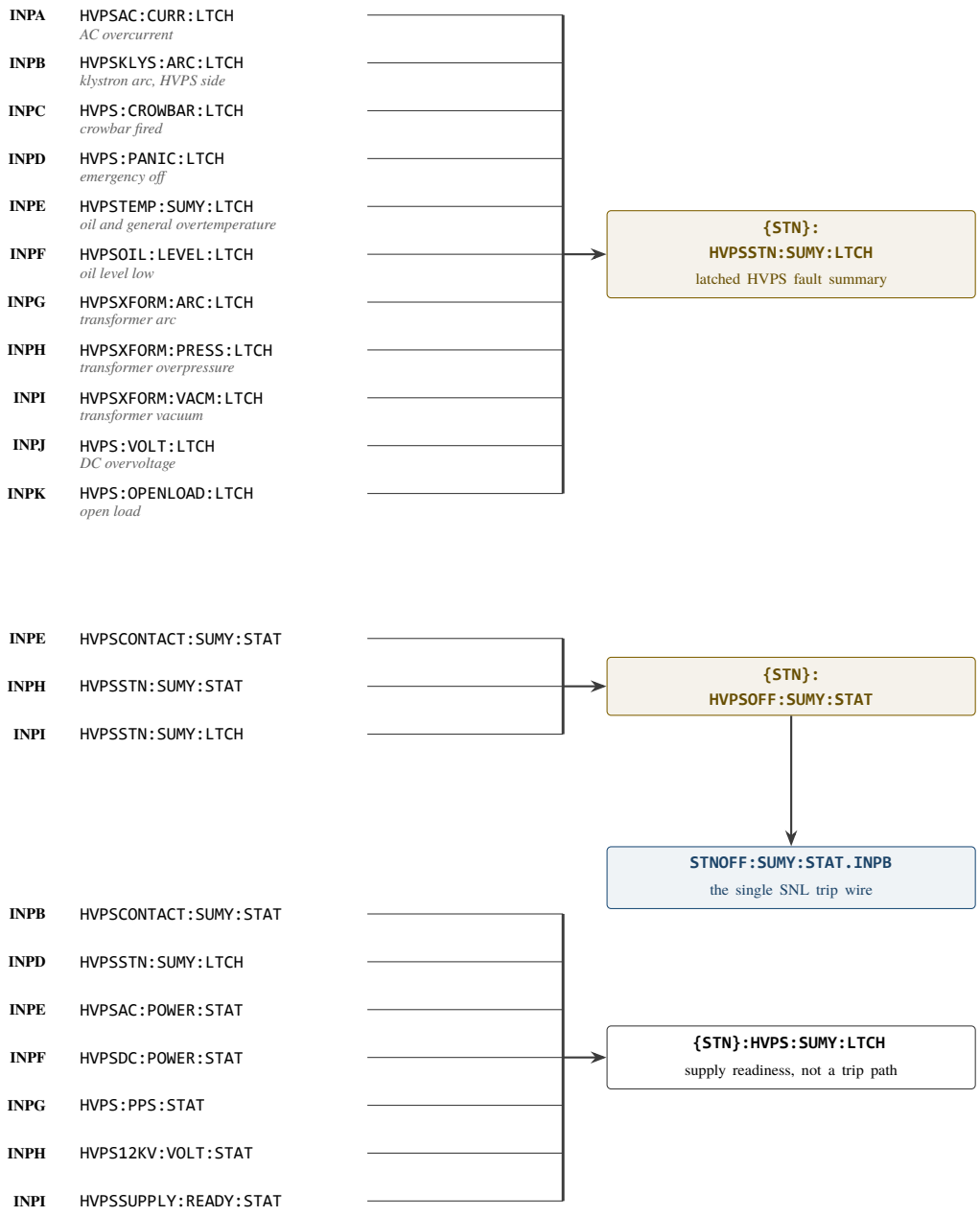


Figure 6: HVPS alarm aggregation. Three summary records, each collecting its inputs on a spine. `HVPSSTN:SUMY:LTCH` is itself an input to both of the others (as INPI and INPD respectively), so a single latched HVPS fault reaches the station trip wire through `HVPSOFF:SUMY:STAT`. `HVPS:SUMY:LTCH` is a readiness summary and is *not* a trip path.

```
    → MX relay (external operational enable)
    → RR relay (reset latch)
    → L1 holding coil
    → Ross Engineering HQ3 vacuum contactor
      (connects/disconnects 12.47 kV AC to HVPS primary)
```

7.8.2 Chain 2 — Ross Grounding Switch

```
PPS 2 Enable (GOB1208PNE Pin G→H)
    → SLC-500 PLC Slot-6 IB16 Input 15
    → PLC Rung 0016 (Ross switch enable)
    → Slot-2 IO8 OUT3 (120 VAC)
    → TS-6 → Belden 83709 cable → Termination Tank
    → Ross Grounding Switch coil

Readback: Ross Switch NC Aux → TS-6 pins 11,12 → GOB1208PNE Readback C-D
```

Critical design issue: The PPS chain routes through SLC-500 ladder logic — a programmable device is in the personnel safety chain. This does not meet modern PPS standards (SLAC ES&H). Flagged in `tex/docL-legacy-architecture/L_legacy_system_architecture.pdf` (identified PPS issues).

7.8.3 Fail-Safe Directions

Both chains are designed so that removing electrical power drives them to the personnel-safe state:

- **Chain 1** (contactor): Operator/PPS removes PPS 1 → K4 relay de-energizes → holding coil L1 loses drive → vacuum contactor spring-opens → 12.47 kV AC is removed from the HVPS primary transformer. A power loss anywhere in the relay chain produces the same outcome.
- **Chain 2** (grounding switch): Operator/PPS removes PPS 2 → PLC Rung 0016 no longer energizes 120 VAC output → Ross switch coil de-energizes → switch spring-closes → HVPS HV bus is grounded. A power loss to the PLC or 120 VAC supply produces the same outcome.

Both readbacks use **NC (normally-closed) auxiliary contacts**, meaning the readback circuit is closed (complete) only when the device is at its safe-access state. A broken wire or failed contact reads as ”unsafe” — fail-safe readback.

7.8.4 Safe-Access Operational Sequence

Personnel access to the SSRL tunnel requires both chains in their safe state **in the correct order**:

```
SHUTDOWN for personnel entry:
  1. PPS 1 removed → Rung 0017 drops → contactor opens → 12.47 kV disconnected
  2. HVPS energy dissipates (capacitor discharge through load)
  3. PPS 2 removed → Rung 0016 drops → Ross switch closes → HV bus grounded
  4. Readbacks A-B and C-D loop complete → PPS confirms both chains safe
  5. Personnel entry permitted

RESTORE after personnel exit:
  1. Personnel withdraw; PPS confirms exclusion at entry points
  2. PPS 2 restored → Rung 0016 energized → Ross switch opens (HV bus un-grounded)
  3. PPS 1 restored → Rung 0017 energized → K4 → contactor closes → 12.47 kV restored
  4. Normal HVPS startup sequence may proceed
```

Critical: The contactor (Chain 1) must be OPEN and HVPS fully discharged **before** the grounding switch (Chain 2) closes onto the HV bus. Closing a grounding switch onto a live high-voltage bus would produce a catastrophic high-energy arc fault. The site PPS system enforces correct sequencing at the machine level through PPS permitting logic.

7.8.5 Compliance Comparison

	Chain 1	Chain 2
Compliance issue	PPS 1 routes through SLC-500 Rung 0017	PPS 2 routes through SLC-500 Rung 0016
Hardware fail-safe	**Present** — OX8 relay input side is wired directly to PPS 1 signal (24VDC from GOB1208PNE Pin E). Even if the PLC closes the output contact, K4 coil cannot energize without PPS 1 voltage physically present.	**Absent** — PLC drives 120 VAC directly to Ross switch coil via IO8 OUT3. If PLC fails energized (maintains 120 VAC output despite PPS 2 removal), the Ross switch remains held open (un-grounded) without a PPS 2 command.
Failure mode concern	PLC closes OX8 spuriously → K4 cannot energize (no PPS 1 voltage) → **no hazard**	PLC maintains 120 VAC spuriously → Ross switch stays open → **HV bus not grounded when PPS removed**

Chain 2 is the higher-priority compliance concern. The spring-return mechanism provides a fail-safe against total power loss, but does not protect against a PLC output failure that maintains the coil energized. The upgrade architecture (see `pps/HoffmanBoxPPSWiring.docx`) eliminates this by providing direct PPS control of both chains through the Interface Chassis, bypassing the PLC for safety-critical permitting.

Sources: [R25] (detailed wiring); `pps/HoffmanBoxPPSWiring.docx`.

---

8 SNL State Machine (`rf_states.st,v`)

Sources: [R2], [R11], [R12], [R22].

8.1 Inputs

Input PV	Source	Connection	Role
`{STN}:STNOFF:SUMY:STAT.SEVR` (`fault_stnoff`)	EPICS alarm aggregation tree (`rf_sumy_stn.db`)	EPICS CA `pvMonitor`	**Master trip wire** — any MAJOR alarm propagated here triggers `s_go_off`
`{STN}:HVPS:VOLT` (`hvpsvoltrd`)	SLC-500 AI (RIO word 30)	EPICS CA `pvGet`	HVPS voltage readback for ramp control

Input PV	Source	Connection	Role
{STN}:HVPS:CURR	SLC-500 AI (RIO word 31)	EPICS CA pvGet	HVPS current readback
{STN}:STN:AIM: ARCLTDSTT	AIM ARCLTDSTT register	EPICS CA pvMonitor	Arc latched status — used to distinguish arc fault from other faults
{STN}:STN:FORCED:LTCH (forced_fault)	AIM fast interlock status register bit	EPICS CA pvMonitor	Forced fault set from EPICS — blocks auto-reset
{STN}:HVPSOFF:SUMY: STAT	EPICS HVPS alarm tree	EPICS CA pvMonitor	HVPS off/fault status, feeds fault_stnoff
{STN}:STN:LOCAL:ON	Local panel switch	EPICS CA	Local/remote mode; local mode forces OFF
{STN}:STN:STATE (stnstate)	EPICS state PV	EPICS CA	Station state (restored at boot in s_init)
{STN}:HVPSSCR:ON:CTRL readback	Remote I/O readback from SLC-500	EPICS CA	SCR enable relay state confirmation
TAXI/GVF status (via rf_msgs.st,v)	GVF module software PVs	EPICS inter-IOC	TAXI error monitoring; triggers resync if TAXI errors detected

8.2 Outputs

Output PV	Destination	Action
{STN}:STN:AIM: FRCBMABT (fba)	AIM MODU.FBA → RF Local Display Panel (output J6 not cabled)	Force Beam Abort — station not ready for beam; set in s_go_off Step 1
{STN}:STN:AIM:MODU. RBA (rba)	AIM MODU.RBA	Reset Beam Abort — de-asserts the forced beam abort during recovery
{STN}:STN:AIM:MODU. RSTF (rstf)	AIM hardware latches	Resets all AIM arc latches and fast interlock latches
{STN}:STN:AIM:MODU. HVPS (aimon)	AIM HVPS_On → RF Local Display Panel (HVPS on request)	Hardware gate for SCR ENABLE; set in HVPSONSUB
{STN}:HVPSSCR:ON: CTRL (hvpstrig)	SLC-500 Remote I/O → Slot 5 OX8 output 0 → trigger boards	Supervisory SCR ENABLE gate; ON in HVPSONSUB, OFF in s_go_off Step 5
{STN}:HVPS:VOLT:SP (hvpswdefault)	SLC-500 Remote I/O → Analog output → SCR firing angle	HVPS voltage setpoint; set to 0 in s_go_off Step 3
{STN}:STN:RFP: RFENABLE (rfswitch)	RFP module RF drive enable	RF drive gate; ON in HVPSONSUB, OFF in s_go_off Step 6
{STN}:STN:STATE (stnstate)	EPICS state display PV	Operator state display: OFF/PARK/TUNE/ON_FM/ON_CW
{STN}:STN:INTCOMP (intcomp)	RFP/IQA integrator compensation	Direct feedback integrator; disabled in s_go_off Step 7

Output PV	Destination	Action
`ef_set(ffwrite_ef)`	`ss rf_statesFF` event flag	Triggers fault waveform capture state set in `s_go_off` Step 8

8.3 Architecture

The SNL state machine (`rf_states.st,v`, 2,227 lines, 3 concurrent state sets) is the **software coordination layer**. It does not provide fast protection — by the time `fault_stnoff` changes, all hardware trips have already fired. Its roles are:

1. **Orderly shutdown** — ramp HVPS to zero before removing SCR enable (protects the HV capacitors/diodes from abrupt current interruption)
2. **Fault recording** — capture of 11 signal channel waveforms to circular buffer
3. **Recovery sequencing** — managed restart with `forced_fault` and auto-reset logic
4. **Operator state interface** — OFF/PARK/TUNE/ON_FM/ON_CW state display
5. **BATS management** — force/reset beam abort in the correct sequence

8.4 The Single Trip Wire: `fault_stnoff`

Figure 7 traces every fault source onto the one record the state machine watches.

Any input going MAJOR alarm propagates (via EPICS `MS` = Maximize Severity) to `STNOFF:SUMY:STAT.SEVR`. Every ON state checks this:

```
/* In s_on_cw, s_tune, s_on_fm */
when (fault_stnoff != NO_ALARM) {
    fault_detected = 1;
} state s_go_off
```

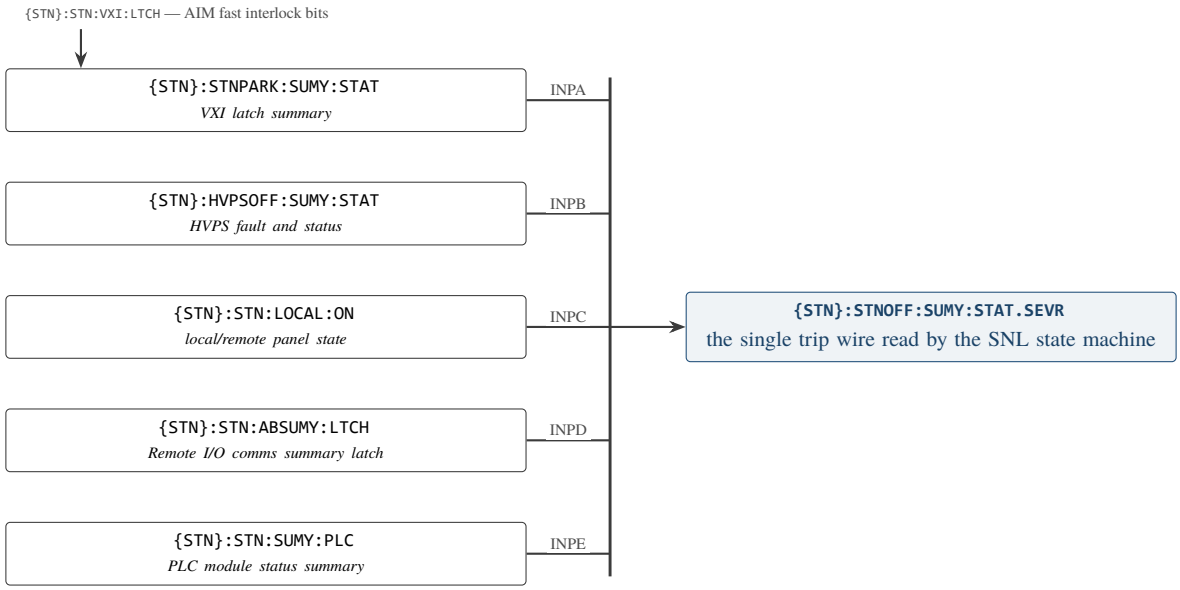


Figure 7: The single SNL trip wire: every fault source converges on `STNOFF:SUMY:STAT.SEVR`.

8.5 STNMPS Alarm Tree (SPEAR3-specific, `rf_sumy_stn_spr.db`)

The SPEAR3-specific MPS summary feeds the station status summary `STN:SUMY:LTCH`, not the trip wire: it is not an input of `STNOFF:SUMY:STAT` [R11], [R12]. Of its inputs, only `STN:VXI:LTCH` also reaches the trip wire, on its own path through `STNPARK:SUMY:STAT`.

```
STN:STNMPS:SUMY:LTCH
    INPB: STN:STN:FORCED:LTCH    (AIM forced fault latch)
    INPC: STN:STN:VXI:LTCH       (AIM latch of RF_FAULT; trips via STNPARK)
    INPE: STN:STN:NCV:PLC        (no-current-value PLC)
    INPF: STN:STNREF:POWER:LTCH  (476 MHz reference power)
    INPG: STN:STNCRATEPS:TEMP:LTCH (VXI crate power supply temperature)
    INPH: STN:STN:RF476MHZREF:LTCH (476 MHz reference status)
    INPI: STN:STN:MPS:LTCH       (External MPS: AIM spare channel, Slot 5 J3)
```

8.6 Orderly Shutdown Sequence (`s_go_off`)

```
State s_go_off (entered when fault_stnoff != NO_ALARM in any ON state):

Step 1: TUNESUB()
    fba = 1   pvPut(fba)        → STN:STN:AIM:FRCBMABT = 1
                                   → MODU.FBA asserted → VXI beam abort (beam abort)
    runmode = TUNE               → puts IQA/RFP in tune mode (low drive)

Step 2: setiqtune()              → zero I/Q drive setpoints

Step 3: pvPut(hvpswdefault, 0)  → write 0 to HVPS voltage setpoint
                                   rf_hvps_loop.st ramps voltage down

Step 4: taskDelay(300)           → ~5 second wait for HVPS to discharge

Step 5: hvpstrig = OFF   pvPut(hvpstrig)
    → STN:HVPSSCR:ON:CTRL = 0  → Remote I/O → SLC-500 (RIO adapter 1)
    → SCR gate drive disabled from supervisory path

Step 6: rfswitch = OFF   pvPut(rfswitch)
    → STN:STN:RFP:RFENABLE = 0 → RF drive enable explicitly removed

Step 7: intcomp = OFF    pvPut(intcomp)
    → Direct feedback integrator compensation disabled

Step 8: efSet(ffwrite_ef)        → triggers ss rf_statesFF (fault file capture)

Step 9: → state s_off
```

8.7 HVPS Startup Sequence (HVPSONSUB macro)

```
/* executed inside s_go_on_cw when transitioning to ON_CW */
HVPSONSUB():
    rfswitch = ON          pvPut(rfswitch)   → RF drive enabled
    pvGet(hvpswdefault)                       → read default voltage setpoint
    /* call TUNESUB to home tuners */
    hvpstrig = ON          pvPut(hvpstrig)   → SCR gate drive enabled via Remote I/O
    aimon = ON             pvPut(aimon)       → write {STN}:STN:AIM:MODU.HVPS = 1
                                                  → AIM fiber "HVPS_On" asserted
    /* wait up to 10 s checking fault_stnoff */
```

The aimon write is a **hardware interlock gate**: the AIM module physically holds its "HVPS_On" output LOW until the IOC writes a 1 to MODU.HVPS. The signal reaches the RF Local Display Panel as the HVPS on request, an input of the logic that forms SCR ENABLE [R26], [R27], so the HVPS cannot be energized unless the software state machine has explicitly authorized it.

8.8 BATS Reset (RESET_BMABTSUB)

```
/* RESET_BMABTSUB(fault) --- called during recovery */
if (fault == NO_ALARM) {
    fba = 0;    pvPut(fba)    → de-assert Force Beam Abort
    pvPut(rba)                 → Reset Beam Abort (MODU.RBA)
    pvPut(rstf)                → Reset Faults (MODU.RSTF --- clears all arc latches)
}
```

8.9 forced_fault Latch — Blocking Auto-Reset

{STN}:STN:FORCED:LTCH is the AIM's forced fault, which an operator sets from EPICS (STN:AIM:FRCFLT) [R7]. When this latch is set:

1. {STN}:STNMPS:SUMY:LTCH.INPB is MAJOR (station status summary, not the trip wire)
2. In s_go_stn_reset (auto-reset state):

```
when (forced_fault != NO_ALARM) {
    /* exit without reset attempt */
} state s_off
```

3. The station stays locked in s_off until an operator turns the forced fault off ({STN}:STN:AIM:FRCFLT) and resets the AIM from EPICS (MODU.RSTF) — this clears the AIM latch and de-asserts forced_fault. The Fast Interlock Chassis has no reset button: its FAULT RESET is a lamp that shows the AIM's reset command [R32]

This was a SPEAR3-specific customization (Sass comment 2004: *"SPEAR is using forced fault to trip the station under certain conditions"*): an operator sets the forced fault from EPICS to trip the station and keep it from resetting itself.

---

9 Fault Event Timeline

9.1 Example: Waveguide Arc

t = 0 ns Arc breakdown at cavity waveguide window:
→ fiber-optic light pulse arrives at Fast Interlock Chassis

t < 1 μs The Fast IC passes the peak-detected pulse; the AIM's comparator fires:
→ the AIM latches the channel (ARCLTDSTT)
→ If the AIM's fault logic drops the HVPS on request or forces
the beam abort, the RF Local Display Panel logic removes SCR ENABLE
(fiber → HVPS controller, B118) and, as its equations specify,
fires the crowbar (O4); HV output then begins to collapse

t < 1 μs AIM hardware asserts RF_FAULT on VXI P2 backplane:
→ RFP module (Slot 4) monitors RF_FAULT pin directly
RF drive DAC output zeroed in hardware
RF power into klystron collapses

t ≈ 5–50 μs AIM ISR fires:
→ Reads ARCLTDSTT latch register (not ISR register — retains fault)
→ scanOnce({STN}:STN:AIM:MODU)

t ≈ 50–500 μs EPICS IOC processes AIM:MODU record:
→ FANO1/2/3 fanout updates ARCCURSTT, ARCLTDSTT, FSTFLT records
→ {STN}:STN:VXI:LTCH goes MAJOR
→ {STN}:STNPARK:SUMY:STAT propagates (MS) → STNOFF:SUMY:STAT MAJOR

t ≈ 10–15 ms MPS PLC (PLC-5/1771) next scan cycle:
→ Detects the fault through its own inputs, if any, and sets its
AB summary fault (fiber → AIM); the collector summary follows
after 0.70 s if the RF cut keeps the collector power
above its limit (§5)
→ Remote I/O: STN:SUMY:PLC = FAULT if it tripped

t ≈ 20 ms SLC-500 detects HVPS crowbar event, if the crowbar fired:
→ Updates HVPS:CROWBAR:LTCH = FAULT in Remote I/O registers
→ HVPSSTN:SUMY:LTCH → HVPSOFF:SUMY:STAT → STNOFF:SUMY:STAT

t ≈ 1 s SNL rf_states.st detects fault_stnoff != NO_ALARM:
→ fault_detected = 1
→ Transitions from s_on_cw → s_go_off state

t ≈ 1.0–1.1 s s_go_off Step 1: TUNESUB()
→ fba = 1 (beam abort forced from the software side)
→ runmode = TUNE

t ≈ 1.1–6 s s_go_off Steps 2–4: HVPS voltage setpoint → 0
→ rf_hvps_loop.st ramps HVPS voltage down to zero
→ ~5 s wait for capacitor bank to discharge

t ≈ 6 s s_go_off Steps 5–7:
→ hvpstrig = OFF (SCR supervisory enable removed via Remote I/O)
→ rfswitch = OFF (RFP drive enable removed)
→ intcomp = OFF

t ≈ 6–10 s ss rf_statesFF: Fault file capture
→ 11 signal channels × /dat/FAULT*_N (N = 1–15 circular)
→ Max 3.6 s for capture to complete

t > 10 s State machine in s_off; operator notified via EPICS alarm

9.2 Example: RF MPS PLC Trip — Cooling Water Overtemperature

This example illustrates a **slow equipment-protection fault** picked up by the RF MPS PLC (PLC-5/1771). No arc occurs — the PLC is the first actor. The cooling summary is not an input of the RF Local Display Panel logic [R27], so high voltage is removed through the RF cut and the collector-power cascade.

t = 0 s Cooling water return temperature sensor reads at scan interval:
→ Exceeds maximum setpoint continuously for N scan cycles
(debounce / confirmation prevents spurious trips)

t ≈ 10–30 ms RF MPS PLC scan cycle completes:
→ PLC program sets cooling overtemperature fault flag
→ AB summary fault set (Path A):
24 V line → RF Local Display Panel → fiber → Fast IC → AIM

t ≈ 10–30 ms AIM responds to the AB summary fault:
→ Asserts RF_FAULT on VXI P2 backplane (open-collector)
→ RFP module (Slot 4) monitors RF_FAULT
RF drive DAC output zeroed in hardware
RF power into klystron collapses

t ≈ 5–50 µs AIM ISR fires:
→ Reads FISTAT latch register (Sass 2004 fix)
→ scanOnce({STN}:STN:AIM:MODU) scheduled

t ≈ 10–30 ms PLC global interlock summary in its Remote I/O data table (Path B):
→ AB-6008 scanner reads at ~1 Hz
→ {STN}:STN:SUMY:PLC → MAJOR alarm set

t ≈ 50–500 ms EPICS propagates alarm:
→ {STN}:STN:ABSUMY:LTCH and {STN}:STN:SUMY:PLC = MAJOR
→ {STN}:STNOFF:SUMY:STAT.SEVR = MAJOR

t ≈ 0.7 s Collector-power cascade, if the collector power has stayed above 1 200 kW for 0.70 s (§5):
→ PLC drops the collector summary
→ RF Local Display Panel logic removes SCR ENABLE
(fiber → HVPS controller, B118); HV output begins to collapse

t ≈ 1 s SNL rf_states.st detects fault_stnoff != NO_ALARM:
→ Transitions s_on_cw → s_go_off

t ≈ 1.0–6 s s_go_off orderly shutdown:
→ Step 1: fba = 1 (beam abort forced from the software side)
→ Steps 2–4: HVPS voltage setpoint → 0; ~5 s ramp
(HVPS already off if the cascade fired)
→ Step 5: hvpstrig = OFF (SCR supervisory enable removed)
→ Step 6: rfswitch = OFF (RF drive enable removed)

t ≈ 6–10 s ss rf_statesFF: Fault file capture
→ 11 signal channels → /dat/FAULT*_N files

t > 10 s State machine in s_off
→ EPICS operator display shows the PLC trip: STN:SUMY:PLC = FAULT
→ No forced_fault latch set (cooling fault, not arc)
→ Auto-reset allowed once fault clears (cooling temp returns to normal)
→ Operator may allow auto-recovery or command manual restart

Key features of this type of trip: (1) The PLC is the first actor (~10–30 ms). (2) RF is cut through the AB summary and the AIM; the HVPS follows through the collector-power cascade into the RF Local Display Panel logic (0.7 s) or the SNL, because the cooling summary is not an input of that logic. (3) No `forced_fault` latch is set, so SNL auto-reset may proceed after cooling fault clears. (4) The fault is identifiable by `{STN}:STN:SUMY:PLC = FAULT` and the PLC's latch for the tripped input, with no arc channel latched in `ARCLTDSTT`.

9.3 Example: HVPS PLC Trip — High-Voltage Transformer Arc

This example illustrates a **HVPS internal fault** monitored by the SLC-500 HVPS PLC. The fault occurs inside the high-voltage transformer/oil tank at B514 — no RF waveguide arc, no MPS PLC involvement.

t = 0 ms Arc discharge inside the HV transformer tank (B514):
→ Arc detector sensor inside HV tank registers arc current surge
→ If arc is energetic enough: crowbar may self-fire via
analog crowbar trigger circuit (independent of HVPS PLC)

t ≈ 10–20 ms SLC-500 HVPS PLC scan cycle (B118):
→ Reads arc detector digital input:
Slot 6 IB16 input: C14.10 = HVPSXFORM:ARC:LTCH = FAULT
→ PLC ladder logic sets HVPS fault state
→ Slot 5 OX8 output 0 (24 V fault/enable) dropped
(supervisory thyristor-trigger enable removed on the COMMANDS bus)
→ Remote I/O output data table updated with fault status

t ≈ 10–20 ms Thyristor triggers inhibited by the SLC-500 supervisory path:
→ HV output begins to collapse (no gate drive on SCR thyristors)
→ HVPS STATUS fiber goes dark (it follows the 24 V fault/enable)
→ RF Local Display Panel receives it as the HVPS beam abort:
its logic latches the beam abort; its effect on SCR ENABLE
and the crowbar is set by the undocumented equations (O4)
→ The STATUS fiber does not reach the Fast IC or the AIM

t ≈ 100 ms–1 s EPICS alarm propagation:
→ AB-6008 scanner reads Remote I/O at ~1 Hz
→ {STN}:HVPSXFORM:ARC:LTCH = MAJOR
→ {STN}:HVPSSTN:SUMY:LTCH.INPG = MAJOR (transformer arc links in)
→ {STN}:HVPSOFF:SUMY:STAT = MAJOR
→ {STN}:STNOFF:SUMY:STAT.SEVR = MAJOR (via INPB)

t ≈ 1 s SNL rf_states.st detects fault_stnoff != NO_ALARM:
→ Transitions s_on_cw → s_go_off

t ≈ 1.0–1.1 s s_go_off Step 1: TUNESUB()
→ fba = 1 (beam abort forced from the software side)
→ runmode = TUNE

t ≈ 1.1–6 s s_go_off Steps 2–4: HVPS voltage setpoint → 0
→ rf_hvps_loop.st attempts HVPS voltage ramp down
(HVPS is already de-energized via the SLC-500 output;
ramp command is a no-op but the sequence must complete)
→ ~5 s wait

t ≈ 6 s s_go_off Steps 5–7:
→ hvpstrig = OFF (SCR supervisory enable again confirmed OFF)
→ rfswitch = OFF (RF drive enable removed)
→ intcomp = OFF

t ≈ 6–10 s ss rf_statesFF: Fault file capture
→ Waveform data captured to /dat/FAULT*_N files
NOTE: Fault occurred in HVPS tank at B514; signal content
in fault files shows the RF state at the time of HVPS collapse

t > 10 s State machine in s_off
→ Operator alarms: HVPSXFORM:ARC:LTCH and HVPSSTN:SUMY:LTCH
→ HVPS must be inspected before restart
→ No arc in RF waveguide: no channel latched, FSTFLT = 0
→ No forced_fault latch (this is HVPS path, not Fast IC/AIM arc)
→ Recovery requires: clearing HVPS arc latch, verifying HVPS
oil/insulation condition, then normal restart sequence

Key features of this type of trip: (1) The SLC-500 HVPS PLC is the first actor (~10–20 ms); the RF MPS PLC and Fast IC are not involved. (2) HVPS shutdown is via the **SLC-500 supervisory output** (§7.6 path) — this is distinct from the SCR ENABLE fiber of the RF Local Display Panel. (3) The HVPS STATUS fiber goes dark with the 24 V fault/enable; the RF Local Display Panel receives it as the HVPS beam abort and sets its beam-abort output, which is not cabled; the fiber does not reach the Fast IC or the AIM. (4) **RF drive shutdown signal chain**: There is no documented hardware path from the SLC-500 HVPS PLC trip to the RFP module RF drive. The RF drive remains asserted at the software level until `s_go_off` Step 6 writes `rfswitch = OFF` at approximately t ≈ 6 s after the initial trip. During the intervening period the klystron cathode HV has already collapsed — the klystron produces no RF output even with drive signal still applied, because klystron amplification requires cathode high voltage. This is safe but the indirection must be understood. See §10.6 for the complete RF shutdown signal trace for this scenario. (5) Fault files capture the RF/IQA signal waveform at HVPS collapse time. (6) The `HVPSXFORM:ARC:LTCH` bit identifies the source; no channel latched in `ARCLTDSTT` and `FSTFLT = 0` confirm no waveguide arc was involved.

9.4 Recovery Sequence

```
Operator observes fault via EPICS display
    → Checks fault files (/dat/FAULT*_N for latest fault slot)
    → Identifies fault channel (ARCLTDSTT bit, FSTFLT word, etc.)

Reset (EPICS):
    → Forced fault off; fault reset to the AIM (MODU.RSTF)
      → Clears the arc latches in the AIM module
      → Clears forced_fault latch
      → ARCLTDSTT bits cleared
      → STN:STN:FORCED:LTCH → OK → STNMPS:SUMY:LTCH clears

EPICS action:
    → fault_stnoff returns to NO_ALARM when all faults clear
    → Operator commands station to ON via EPICS display

SNL response:
    → s_off → s_go_park → s_go_tune → s_go_on_cw
    → HVPSONSUB():
        rfswitch = ON
        hvpstrig = ON  → SLC-500 (RIO adapter 1) output 0 on → thyristor triggers enabled
        aimon = ON     → AIM "HVPS_On" asserted → HVPS on request to the panel
    → RESET_BMABTSUB():
        fba = 0 → rba → rstf  (beam abort reset, arc hardware latches cleared)
    → Station climbs to ON_CW
```

9.5 Example: SPEAR MPS Permit Withdrawal — Cascade to Collector Overpower

This example traces the complete fault event when SPEAR MPS withdraws its beam permit (e.g., due to a beam loss or machine protection interlock condition). It illustrates the Slot 5 primary RF cut followed by the collector overpower cascade described in §5.

System state before fault: RF station ON_CW, HVPS at about 72 kV and 19.4 A (cathode power about 1.4 MW, the June 2020 operating point [R1]), normal beam circulating in SPEAR3.

t = 0 ms SPEAR MPS withdraws beam permit:
→ Permit input of the Slot 5 MPS Shutoff module drops
→ Slot 5 asserts RF_FAULT on VXI P2 backplane
→ RFP module (Slot 4) RF drive DAC → 0 (hardware, < 1 μs)
→ Slot 5 J3 → Fast IC spare-fault input → AIM External MPS
→ Klystron RF output: begins dropping to ~0 (~1–5 μs settling)

t ≈ 10 μs AIM Module (Slot 12) ISR fires:
→ STN:VXI:LTCH = MAJOR; STN:MPS:LTCH = FAULT (External MPS)
→ STNPARK:SUMY:STAT alarm begins propagating through EPICS tree
At this instant
HVPS still fully energized at ~72 kV, ~19 A cathode current
RF output ≈ 0; cavity begins ring-down (~1 ms time constant)
ALL cathode beam power now depositing in klystron collector
P_collector ≈ P_cathode (≈ 1.4 MW, above the 1200 kW limit)

t ≈ 0.7 s RF MPS PLC collector protection:
→ Collector power (HV × current − RF output) above 1200 kW
since t = 0; the PLC-5 timer runs out after 0.70 s
→ PLC latches collector overpower and drops its collector
summary and its global interlock summary → Path A activated:

t ≈ 0.7 s Path A — RF Local Display Panel logic responds (μs scale):
→ SCR ENABLE fiber → HVPS controller (B118) goes dark
→ thyristor triggers inhibited
→ HVPS HV output begins to collapse
→ KLYSTRON CROWBAR fiber dark if the logic so specifies (O4)
→ crowbar discharges the residual capacitor energy
→ AB summary fiber → AIM → RF_FAULT (redundant RF cut confirmation)
Remote I/O latch: KLYSCOLL:POWER:LTCH (WL=16/WF=31/B=2) = MAJOR latched
(hardware latch confirming the PLC trip; propagates to EPICS)

t ≈ 0.7 s HVPS STATUS fiber goes dark → RF Local Display Panel (HVPS beam abort)
[the panel's beam-abort output is already set — see §3.8]

t ≈ 1 s EPICS Remote I/O data arrives at IOC (~1 Hz):
→ {STN}:KLYSCOLL:POWER:LTCH propagates → KLYS:SUMY:LTCH = MAJOR
→ STN:SUMY:PLC and STN:ABSUMY:LTCH = MAJOR
(STNOFF:SUMY:STAT is already MAJOR through STN:VXI:LTCH)

t ≈ 1 s SNL s_go_off sequence begins:
→ Step 1: fba = 1 (beam abort forced from the software side)
→ Steps 2–4: HVPS ramp setpoint → 0 and hvpstrig=OFF
[HVPS already hardware-killed at ~0.7 s if the cascade fired]
→ Step 5: rfswitch = OFF (software confirms RF drive off;
was already 0 from Slot 5 at t = 0)
→ Step 6: Fault files captured (11 RF/IQA channels)
Station is in s_fault/s_off state.

EPICS fault signature:

PV	Value	Interpretation
`{STN}:STN:VXI:LTCH`	MAJOR	Slot 5 RF_FAULT seen by AIM ISR (root cause — timestamps ~10 μs after t=0)

PV	Value	Interpretation
{STN}:STN:MPS:LTCH	FAULT	External MPS: the AIM's spare channel, driven by Slot 5 J3 (root cause)
{STN}:KLYSCOLL:POWER:LTCH	MAJOR	Collector overpower trip in the PLC (cascade precipitating HVPS kill — about 0.7 s after t=0)
{STN}:STN:AIM:ARCLTDSTT	0x70FB	Bit 8 (External MPS) latched; every arc channel clear — not an arc event
{STN}:STN:ABSUMY:LTCH	MAJOR	AB summary fault from the RF MPS PLC (collector overpower)
{STN}:HVPS:VOLT	falling from about 0.7 s	Thyristor triggers inhibited by the panel's SCR ENABLE (cascade), or the SNL ramp from about 1 s

Key feature: The KLYSCOLL:POWER:LTCH flag appears as the hardware event that killed the HVPS — even though the root cause was SPEAR MPS. An operator must compare EPICS archiver timestamps: STN:MPS:LTCH timestamp precedes KLYSCOLL:POWER:LTCH by about 0.7 s, confirming the cascade direction. With default 1 Hz archiver resolution, both may appear in the same or adjacent samples; event-driven archiving is required for precise sequencing.

Recovery: Standard recovery sequence (§9.4). SPEAR MPS must restore its permit; a fault reset then clears STN:VXI:LTCH and STN:MPS:LTCH in the AIM and, through the PLC-5 reset, KLYSCOLL:POWER:LTCH, before the station can return to ON_CW.

---

10 Key Cross-Layer Interactions

10.1 AIM aimon ↔ SCR ENABLE Hardware Gate

The aimon write ({STN}:STN:AIM:MODU.HVPS = 1) sets the AIM "HVPS_On" output, which reaches the RF Local Display Panel as the HVPS on request. This is a **hardware interlock gate** — the panel's logic forms SCR ENABLE from it, so the HVPS controller in B118 cannot fire its thyristors unless the AIM has asserted it [R26], [R27]. This cross-link means:

- The SNL state machine must explicitly authorize HVPS energization (via HVPSONSUB)
- If the software is not in the ON transition sequence (e.g., the IOC is dead), AIM holds this output LOW and HVPS cannot be energized

In the fault path, aimon is NOT explicitly cleared in s_go_off — the hardware trip (SCR ENABLE removed by the panel logic + SLC-500 hvpstrig=OFF) has already shut down the HVPS. The explicit aimon=ON is only needed at startup.

10.2 forced_fault Anti-Auto-Reset

Without {STN}:STN:FORCED:LTCH, the auto-reset state (s_go_stn_reset) would retry reset_count times automatically. With forced_fault = MAJOR, the auto-reset loop exits immediately to s_off, requiring operator intervention. This keeps a station that an operator has tripped on purpose from re-energizing itself.

10.3 VXI Interlock Latch Register Fix (2004)

Before January 2004, VXI interlock bits were read from the AIM interrupt status register (ISR). EPICS processed the AIM:MODU record continuously, and each read cleared the ISR — a latched fault could be

invisible to the alarm system because it self-cleared between EPICS scans. Sass changed `rf_interlock_vxi.db` (rev 1.2, Jan 2004) to read from the AIM **fast interlock status/latch register**, which holds bits until a fault reset clears them. **This is a critical reliability fix** — without it, the software alarm chain would miss fast faults.

10.4 Dual SCR Enable Paths — Supervisory vs. Fast

The HVPS SCR bank has two independent enable paths that operate in different domains:

Path	Owner	Speed	Set by	Cleared by
RF Local Display Panel → SCR ENABLE fiber → trigger boards (B118)	Hardware interlock	μs	HVPS on request with no fault	Panel logic (O4)
SLC-500 output 0 → COMMANDS bus (B118)	Supervisory enable	~20 ms	SNL `hvpstrig=ON` in `HVPSONSUB`	SNL `hvpstrig=OFF` in `s_go_off`

The SLC-500 output is the **supervisory gate** — it deliberately enables the HVPS at the beginning of the startup sequence and is the controlled OFF path during orderly shutdown. The panel's SCR ENABLE fiber is the **fast-trip gate** — held lit during normal operation, dark on fault. Both must be satisfied simultaneously for the HVPS to produce HV output.

10.5 RF MPS PLC Two-Path Architecture — Defense in Depth

A single RF MPS PLC fault produces consequences on two independent signal paths simultaneously (see §6.5 for full description). This is a cross-layer interaction worth noting explicitly:

Path	Layer	Immediate HVPS Effect	Delayed HVPS Effect (SNL)	RF Effect	Speed	Notes
A — MPS summaries → panel logic and AIM	Hardware interlock	**SCR ENABLE removed by the panel logic** for collector and waveguide pressure faults (fiber → HVPS controller, B118); crowbar per O4	Orderly `hvpstrig=OFF` (~1 s, HVPS already dead)	RF_FAULT via AIM	< 15 ms total	Primary HVPS shutdown path for the faults wired into the panel logic
B — Remote I/O global interlock summary (`STN:SUMY:PLC`)	Software / EPICS	None	SNL orderly `s_go_off`: HVPS ramp + `hvpstrig=OFF`	SNL orderly shutdown	~1 s	Fault record, fault file, state machine bookkeeping; hardware already acted

For SPEAR MPS permit and orbit interlock (Slot 5 direct inputs):

Input	Immediate HVPS Effect	Delayed HVPS Effect (SNL)	RF Effect	Speed
SPEAR MPS beam permit removal	**None** from Slot 5 — no panel path; the AIM may act on its External MPS channel (O3), and the collector overpower cascade fires Path A after 0.7 s (§5)	Orderly SNL `s_go_off` (~6 s)	RF_FAULT via Slot 5	Not documented (O1)
Orbit interlock trip	**None** from Slot 5 — no panel path; the AIM may act on its External MPS channel (O3), and the collector overpower cascade fires Path A after 0.7 s (§5)	Orderly SNL `s_go_off` (~6 s)	RF_FAULT via Slot 5	Not documented (O1)

The design consequence: for an RF MPS PLC trip on collector power or waveguide pressure, the immediate HVPS hardware shutdown is **owned by Path A** through the RF Local Display Panel logic; other MPS faults reach the HVPS through the collector-power cascade or the SNL. Path B cannot produce an immediate HVPS shutdown. If Path A were to fail (e.g., a panel fault), the SNL would still perform orderly HVPS shutdown ~6 s later through Path B, but there would be no immediate HVPS hardware kill.

For SPEAR MPS and orbit interlock signals, Slot 5 has **no connection** to the SCR ENABLE or crowbar fibers. A hardware HVPS kill can come only from the AIM, if its fault logic acts on the External MPS channel (O3), or from the cascade: with drive = 0 and HVPS on, the klystron collector absorbs full cathode power, and the RF MPS PLC drops its collector summary (Path A) once that power has stayed above 1 200 kW for 0.70 s; the panel logic then removes SCR ENABLE. Otherwise the HVPS is shut down by the SNL, about 6 s after the trip. See §5 for the complete physics and signal chain, and §9.5 for an example fault timeline.

This asymmetry should be considered during upgrade architecture design: the new Interface Chassis interlock design must maintain an equivalent hardware-speed path from MPS signals to the HVPS SCR/crowbar circuit. Although the cascade (§5, §10.7) provides a de facto hardware HVPS kill after about 0.7 s, it relies on the klystron physics and the RF MPS PLC — a **direct** wired path from SPEAR MPS permit removal to the HVPS SCR kill (bypassing the cascade dependency) is currently absent and should be evaluated for the upgrade.

10.6 RF Shutdown Path Following HVPS PLC Trip

Open question resolved. When the SLC-500 HVPS PLC trips and shuts off the HVPS, the RF drive is **not immediately cut** by any documented hardware signal from the HVPS PLC. The SLC-500 has no wired path to the Fast Interlock Chassis or the AIM; its only link to the RF side is the STATUS fiber, which ends on the RF Local Display Panel.

The complete RF shutdown signal trace for a HVPS PLC-only trip is:

`t = 0 ms` SLC-500 HVPS PLC detects internal fault (e.g., transformer arc):
→ Slot-5 output 0 (24 V fault/enable) drops
→ Thyristor triggers inhibited on the COMMANDS bus (B118)
→ HVPS HV output begins to collapse (SCR gates disabled)

`t ≈ 0 ms` HVPS STATUS fiber goes dark (it follows the 24 V fault/enable):
→ RF Local Display Panel receives the HVPS beam abort
→ The panel's beam-abort output is set; it is not cabled, so nothing leads from it to the AIM or the RFP

t ≈ 0 ms KLYSTRON CATHODE VOLTAGE COLLAPSES:
→ Klystron cathode at approximately 0 V (or HVPS residual)
→ RF drive signal still present from RFP module (software unaware)
→ Klystron cannot amplify without cathode HV: RF output = ~0
→ Cavity fields decay on cavity time constant (~1 ms for SPEAR3)

t = 0 ms RF DRIVE SIGNAL PATH (VXI → klystron):
→ RFP module (VXI Slot 4) continues outputting RF drive DAC values
→ RF_FAULT NOT asserted (no AIM arc latch, no AB summary fault)
→ rfswitch PV still = ON

t ≈ 100 ms–1 s EPICS alarm propagation (Remote I/O → AB-6008 scanner → EPICS):
→ {STN}:HVPSXFORM:ARC:LTCH (or other HVPS fault latch) = MAJOR
→ {STN}:HVPSSTN:SUMY:LTCH propagates to HVPSOFF:SUMY:STAT
→ {STN}:STNOFF:SUMY:STAT.SEVR = MAJOR
→ fault_stnoff != NO_ALARM in rf_states.st

t ≈ 1 s SNL rf_states.st detects fault → enters s_go_off:
→ Step 1: fba = 1 (beam abort forced from the software side)
→ Steps 2–4: HVPS voltage setpoint → 0 (no-op; already collapsed)
→ Step 5: hvpstrig = OFF (SCR supervisory enable confirmed OFF)
→ Step 6: rfswitch = OFF
{STN}:STN:RFP:RFENABLE = 0
→ RFP module RF drive DAC output disabled
← RF DRIVE IS CUT HERE (t ≈ 6 s after initial trip)

Why the ~6 s latency is safe: A klystron is a velocity-modulation amplifier that requires cathode high voltage to produce RF output. Without cathode HV, even a full-amplitude drive signal produces no forward power. The cavity fields have completely decayed within ~1 ms of HV collapse. The ~6 s period between HVPS trip and `rfswitch = OFF` therefore does not produce any RF output and poses no equipment hazard — the klystron is a passive, inert component during this window.

No direct hardware RF kill path from HVPS PLC: Unlike an arc at the Fast IC (where the AIM asserts RF_FAULT to cut drive), the SLC-500 HVPS PLC trip path has no documented direct RF kill. Its STATUS fiber ends on the RF Local Display Panel, whose beam-abort output is not cabled.

Possible indirect secondary path (not guaranteed): If the sudden HVPS voltage collapse causes reflected power to rise at the cavity junctions (forward power drops → VSWR increases), the RF MPS PLC may detect a reflected power trip condition and set its AB summary fault (Path A: fiber → AIM → RF_FAULT). This would cut RF drive in ~10–15 ms via AIM → VXI backplane. However, this secondary path is **not guaranteed**: if drive power was low before the trip, reflected power may stay below the MPS threshold.

RF shutdown timeline summary for HVPS PLC trip:

Time	Event	RF Drive State
t = 0	HVPS PLC trips; SCR ENABLE removed	Drive ON; klystron de-powered
t ≈ 0	HVPS HV collapses; klystron cathode at ~0 V	Drive ON (software unaware); RF output = 0
t ≈ 100 ms–1 s	EPICS alarms propagate via Remote I/O	Drive ON (SNL not yet triggered)
t ≈ 1 s	SNL s_go_off sequence begins	Drive ON (Steps 1–5 in progress)
t ≈ 6 s	**s_go_off Step 6: rfswitch = OFF**	**Drive OFF — RF drive cut**

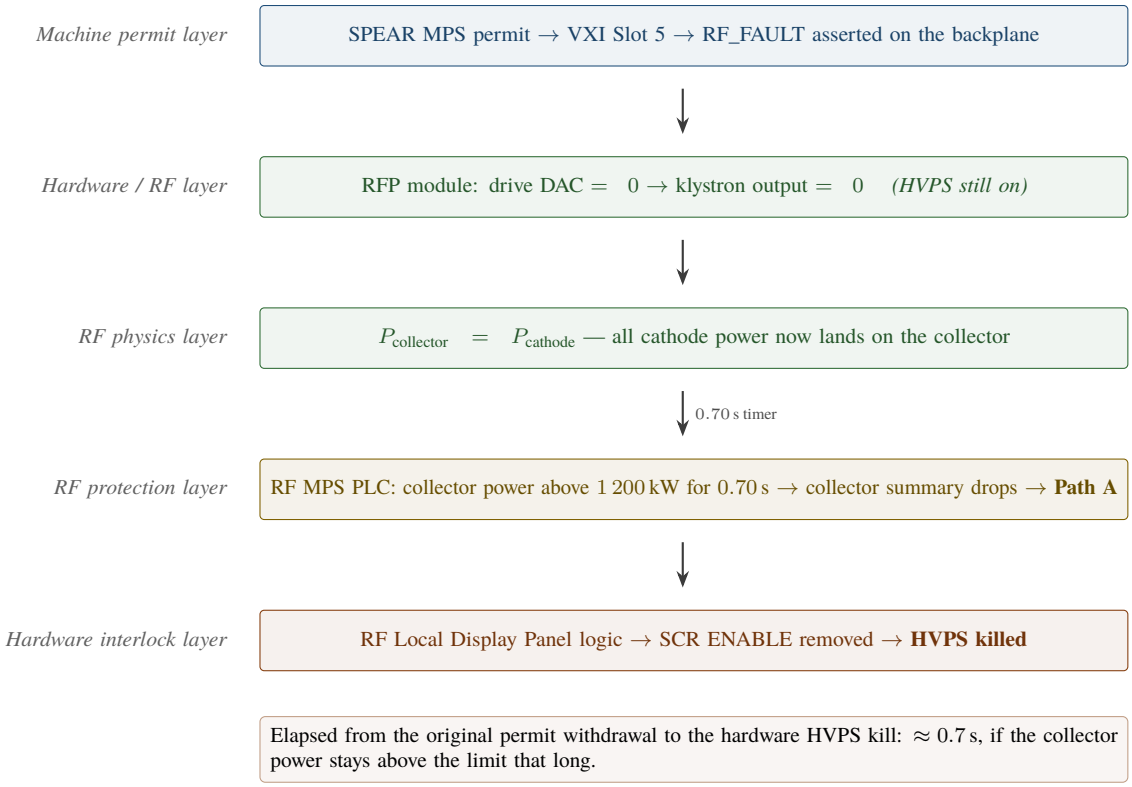


Figure 8: Permit layers. The machine permit and personnel permit chains are separate all the way down.

Design implication for the upgrade: The upgrade Interface Chassis interlock design should include a direct hardware path from HVPS enable removal (or HVPS PLC fault output) to the LLRF9 RF enable input. This would eliminate the ~6 s software-dependent window and make the RF shutdown timing visible and auditable in hardware — consistent with the defense-in-depth philosophy of the rest of the interlock architecture.

10.7 Cascade Cross-Layer Interaction: RF Drive Cutoff → Secondary Trip Chain

The collector overpower cascade (§5) is a cross-layer interaction in which an event that originates in the machine-protection / permit layer (SPEAR MPS permit withdrawal → Slot 5 RF cut) propagates back into the hardware protection layer (RF MPS PLC collector summary → RF Local Display Panel logic), producing a hardware HVPS kill about 0.7 s later.

This behavior is consistent with the defense-in-depth design principle: the RF MPS PLC independently monitors a physical parameter (collector power) that is deterministically correlated with the RF drive state. It does not need to ”know” that a SPEAR MPS event occurred — it simply responds to an out-of-bounds physical condition.

Cross-layer cascade path (Figure 8):

Three key design observations:

1. **The cascade is deterministic** for normally-operating klystrons. RF drive cutoff with HVPS energized will always drive the collector to absorb full cathode power. The trip limit is fixed at 1 200 kW in the PLC-5 program, below the full cathode power at the normal operating point (about 1.4 MW), so the PLC drops its collector summary 0.70 s after the cut unless the high voltage has been lowered first [R37].

2. **Path A is the hardware HVPS kill for collector overpower** — whether triggered by a primary RF MPS PLC protection event or by this cascade, the collector summary into the RF Local Display Panel logic is the mechanism. Any detected collector overpower therefore removes SCR ENABLE in hardware, regardless of how the overpower arose.
3. **Fault diagnosis transparency** — the cascade produces a fault signature dominated by `KLYSCOLL:POWER:LTCH` (collector overpower), while the actual root cause is visible in `STN:MPS:LTCH` and/or `STN:VXI:LTCH`. The fault analysis procedure (§11.6) should explicitly check for this cascade pattern when a collector overpower latch coincides with a machine permit withdrawal:
   - `STN:VXI:LTCH` timestamps $\ll$ `KLYSCOLL:POWER:LTCH` timestamps $\rightarrow$ cascade (root cause is the permit withdrawal)
   - `KLYSCOLL:POWER:LTCH` timestamps with no preceding permit change $\rightarrow$ primary collector overpower event (possible klystron efficiency issue)

Upgrade architecture note: The cascade provides a *de facto* hardware HVPS kill for SPEAR MPS and orbit interlock trips via an indirect physics path. The upgrade Interface Chassis design should consider whether to implement a *direct* hardware path (permit removal $\rightarrow$ HVPS SCR kill, without depending on the klystron physics cascade), which would be faster (~1 ms vs. ~0.7 s), more transparent in fault records, and functional even at low RF power levels where the cascade may not fire.

---

11 Fault Data Availability and Analysis

When a fault event occurs, multiple data sources capture pre- and post-event waveforms at different speeds and with different signal content. This section describes where each data source is stored, how to access it, and provides a step-by-step fault analysis procedure.

11.1 Overview of Fault Data Sources

Source	What It Captures	Time Resolution	Latency to Availability	Storage Location
AIM Hardware History Buffer	12 arc channel voltages + HVPS voltage (continuous ADC ring buffer)	~μs per sample	Immediately (freezes on fault)	VXI AIM module on-board memory; exported to IOC `/dat/aimHist.dat`
SNL Fault Files (`/dat/FAULT*_N`)	11 RF/IQA signal channels (I/Q waveforms, amplitude waveforms)	~1 ms per sample	~6–10 s post-fault	VxWorks virtual disk `/dat/` on VXI IOC
B118 Oscilloscope (4-channel)	HVPS DC voltage, HVPS DC current, inductor T2 sawtooth voltage, transformer T1 AC current	Oscilloscope sweep rate (operator-set; typical 10–100 ms/div)	Continuous display; capture on trigger	Local oscilloscope (memory, USB, or printout)
EPICS Channel Archiver	All EPICS PVs — alarm states, analog readbacks, timestamps, state machine transitions	1 Hz nominal; event-driven on PV change	Always available; rolling multi-year history	EPICS archiver server (see §11.5)

11.2 AIM Hardware History Buffer

What it contains: The AIM's on-board fast ADC continuously digitizes the 12 arc channel voltages from the Fast Interlock Chassis (340-308) and the HVPS DC voltage into the AIM module's 512 KB ring buffer (`HISBUF`, A24 address space, offset `0x0038`). On fault, the `ADCCTL` bit freezes the buffer, preserving the pre-fault arc waveforms.

Where stored: VXI AIM module on-board memory. The device driver (`devP2RfAim.c`) reads arc voltage peaks via the `ARCVOL` register and crate voltage via `CRTVOL`. The history data is exported to `/dat/aimHist.dat` on the IOC. Note: buffer readout is **gated** — if `FISTAT.HVPSON = 0` (HVPS off at fault time), the driver skips arc voltage readback.

EPICS readback PVs (immediately available post-fault):

PV	Content
`{STN}:STN:AIM:ARCCURSTT`	Real-time arc channel status, one bit per channel (1 = clear; a tripped channel reads 0 until the arc ends)
`{STN}:STN:AIM:ARCLTDSTT`	Latched arc channel status, one bit per channel (a latched channel reads 0 until a fault reset)
`{STN}:STN:AIM:FSTFLT`	First-fault arc channel
`{STN}:STN:AIM:INTSTATE`	AIM overall interlock state

Interpreting `ARCLTDSTT` bit pattern (1 = clear, 0 = latched; disabled channels read 0) [R3], [R35]:

Bits	Channel	Latch record
0	Klystron arc	`KLYS:ARC:LTCH`
1	Circulator arc	`CIRC:ARC:LTCH`
3–6	Cavity 1–4 window arc, air side	`CAV`*`n`*`AIR:ARC:LTCH`
7, 12–14	Cavity 1–4 window arc, vacuum side	`CAV`*`n`*`VACM:ARC:LTCH`
8	External MPS: SPEAR MPS or orbit permit, through Slot 5	`STN:MPS:LTCH`
2, 15	Not named; disabled	Always 0
9–11	Not channels	Always 0
`0x71FB`	**Nothing latched** (enables of 11 November 2026)	A latched channel clears its bit

11.3 SNL Fault Files (`/dat/FAULT*_N`)

What they contain: 11 pre-fault RF and IQA signal channels captured from hardware signal memory buffers in the RFP and IQA VXI modules. These represent the RF state at the moment of fault.

Where stored: VxWorks virtual disk `/dat/` on the VXI IOC in B132. Files use the naming convention `FAULT<channel>_<slot>` where `<slot>` is 1–15 (circular buffer, `NUMFAULTS=15`). The current slot number is tracked as EPICS PV `{STN}:STN:FAULT:NUM`.

File channel list (in capture order for SPEAR3 `#else` branch of `#ifdef CF2`):

Slot Index	File Pattern	Signal
0	`FAULTRfpSI_N`	RFP Signal I (cavity forward, I component)
1	`FAULTRfpSQ_N`	RFP Signal Q (cavity forward, Q component)
2	`FAULTRfpCI_N`	RFP Cavity I (cavity voltage, I component)
3	`FAULTRfpCQ_N`	RFP Cavity Q (cavity voltage, Q component)
4	`FAULTIqa1Amp_N`	IQA1 amplitude waveform
5	`FAULTIqa2Amp_N`	IQA2 amplitude waveform
6	`FAULTGvf_N`	GVF signal (PEP-II heritage; inactive in SPEAR3)
7	`FAULTAim_N`	AIM arc/interlock status snapshot
8	`FAULTCmbI_N`	Combiner I (PEP-II; inactive in SPEAR3)
9	`FAULTIqa3Amp_N`	IQA3 amplitude waveform
10	`FAULTCmbQ_N`	Combiner Q (PEP-II; inactive in SPEAR3)

Access procedure:

1. Check `{STN}:STN:FAULT:NUM` EPICS PV to find the most recent fault slot number N
2. Log in to the VxWorks IOC shell (telnet to B132 VXI CPU) or access via NFS/FTP
3. List `/dat/FAULT*_N` (substitute the slot number)
4. Transfer files to a workstation for analysis (binary format; see §11.6 for reading)

11.4 B118 Oscilloscope — Four HVPS Monitor Channels

Location: Building B118, HVPS Hoffman Box equipment area. An oscilloscope is permanently installed to monitor 4 analog signals from the HVPS system.

The four channels are physical analog signals routed from the HVPS cabinet to BNC inputs on the oscilloscope via shielded cables. The oscilloscope is a standalone instrument; it is **not** connected to EPICS. The signal identities are:

Channel	Signal	Source Hardware	Expected Waveform Pattern	Diagnostic Value
CH1	HVPS DC Output Voltage	Resistive voltage divider (≈10,000:1) from HVPS DC bus	Smooth DC at ≈ −72 to −75 kV typical; slight 12-pulse ripple (~6.9% P-P)	Crowbar: instant collapse to ~0 V; SCR disable: decay over ~100 ms; setpoint changes visible as slow ramp
CH2	HVPS DC Output Current	Danfysik DC-CT (Hall-effect current transducer)	DC at ~22 A nominal	Arc event: current spike; crowbar: current spike then zero; HVPS trip: current decay

Channel	Signal	Source Hardware	Expected Waveform Pattern	Diagnostic Value
CH3	Inductor 2 (T2) Sawtooth Voltage	SCR firing circuit monitor winding on T2	**Bipolar asymmetric sawtooth** (inverted direction vs. theoretical model — this is normal for real SPEAR3)	Waveform disappears when SCR firing stops; asymmetry changes indicate firing angle / regulation shifts; absence indicates HVPS off
CH4	Transformer 1 (T1) AC Phase Current	Current transducer on T1 primary winding	**Bipolar asymmetric square pulses** with commutation spikes (not sinusoidal — characteristic of 12-pulse discrete switching)	Dropout indicates HVPS de-energized; asymmetric pulses indicate phase balance issue; commutation spike amplitude indicates SCR health

Accessing data for fault analysis:

- **Field capture**: Use the oscilloscope's single-shot trigger or freeze function at the time of fault; export via USB/Ethernet or photograph the screen. The scope is a standalone instrument — no EPICS waveform interface exists for this unit.

11.5 EPICS Channel Archiver

What it contains: The SLAC EPICS Channel Archiver continuously records all EPICS PVs with timestamps. This provides the complete alarm and measurement history for all PVs in the RF station, typically with years of rolling history.

Access methods:

1. **EPICS Strip Chart / Archive Viewer GUI**: Open the Strip Chart tool from any EPICS operator display. Enter PV name(s) and select the time window around the fault event.
2. **Channel Archiver web interface**: Contact SLAC SSRL controls group for the archiver server hostname and web URL.
3. **Python (Archiver Appliance REST API)**:

```
import requests
pv = "SRF1:STN:AIM:ARCLTDSTT"  # substitute actual station prefix
t0 = "2026-04-10T02:00:00Z"      # start time (UTC ISO-8601)
t1 = "2026-04-10T03:00:00Z"      # end time
url = f"http://<archiver-host>/retrieval/data/getData.json?pv={pv}&from={t0}&to={t1}"
data = requests.get(url).json()
```

Key PVs to retrieve for fault post-mortem:

Category	PV	What to Look For
Arc fault	`{STN}:STN:AIM:ARCLTDSTT`	Which enabled bit(s) cleared at fault time?
First arc channel	`{STN}:STN:AIM:FSTFLT`	Which arc channel tripped first
External MPS	`{STN}:STN:MPS:LTCH`	Did the SPEAR MPS or orbit permit trip the station (Slot 5)?
MPS PLC	`{STN}:STN:SUMY:PLC`	Did the RF MPS PLC trip?

Category	PV	What to Look For
HVPS summary	`{STN}:HVPSSTN:SUMY:LTCH`	Which HVPS sub-fault latched?
Transformer arc	`{STN}:HVPSXFORM:ARC:LTCH`	HV transformer internal arc event?
Crowbar	`{STN}:HVPS:CROWBAR:LTCH`	Crowbar activated?
Overvoltage	`{STN}:HVPS:VOLT:LTCH`	DC output overvoltage?
HVPS voltage	`{STN}:HVPS:VOLT`	Pre-fault and collapse waveform (1 Hz; limited time resolution)
Station state	`{STN}:STN:STATE`	State machine transitions: ON_CW → s_go_off → OFF
Fault slot	`{STN}:STN:FAULT:NUM`	Fault file slot number — use to locate `/dat/FAULT*_N` files
Forced fault	`{STN}:STN:FORCED:LTCH`	Forced fault set from EPICS (`STN:AIM:FRCFLT`); blocks the automatic reset until it is turned off and reset

11.6 Step-by-Step Fault Analysis Procedure

11.6.1 Immediate Response (within minutes of fault)

1. **Record exact UTC timestamp** from the EPICS alarm notification or operator display
2. **Read the EPICS alarm display** on the EDM operator panel — identify which latches are set:
   - An enabled bit of `ARCLTDSTT` at 0 → that channel latched; note which bit(s) (bit 8 = External MPS)
   - `HVPSXFORM:ARC:LTCH` set → HV transformer internal arc
   - `HVPS:CROWBAR:LTCH` set → Crowbar fired
   - `STN:MPS:LTCH` = FAULT → SPEAR MPS or orbit permit lost (Slot 5)
   - `STN:SUMY:PLC` = FAULT → RF MPS PLC trip (cooling, vacuum, or power)
   - `STN:FORCED:LTCH` set → Forced fault from EPICS; turn it off and reset before recovery
3. **Note fault slot number** from `{STN}:STN:FAULT:NUM` — needed to retrieve fault files
4. **Go to B118** and check the oscilloscope display for the fault waveforms:
   - Was CH1 a sudden drop (crowbar) or a slow decay (SCR disable)?
   - Did CH4 drop at the same instant as CH1, or before?
5. **Do NOT issue a fault reset** until the cause is understood — the reset clears the arc latch evidence

11.6.2 Detailed Analysis (after initial survey)

1. **Retrieve fault files** from IOC: `telnet <B132-IOC-hostname>`, then check `/dat/FAULT*_N` for the slot noted above
2. **Retrieve EPICS archiver history** for the key PVs listed in §11.5 in a ±5 minute window around the fault timestamp
3. **Plot fault file data**. The files are binary with a short header followed by raw ADC samples as 16-bit integers or 32-bit floats (format is hardware-dependent; consult `devP2RfAim.c` for AIM fault file format or the RFP/IQA driver for RF signal file format). In Python:

```
import numpy as np
data = np.fromfile('/dat/FAULTRfpSI_1', dtype=np.int16)  # example
```

4. **Review the B118 oscilloscope capture** (see §11.4): note whether CH1 (DC voltage) shows an abrupt collapse (crowbar, < 1 μs) or a gradual decay (~100 ms for SCR disable), and whether CH4 (T1 AC current) dropped simultaneously with or prior to CH1
5. **Determine the initiating event** — use EPICS timestamp ordering (1 Hz resolution for Remote I/O PVs) to sequence events among HVPS faults; use the arc channel bit pattern for hardware-speed (< 1 μs) faults

11.6.3 Fault Categorization Guide

Observed Indicators	Most Likely Initiating Event	First Actor
`ARCLTDSTT` bit 0 or 1 cleared	Klystron or circulator arc	AIM, on the Fast IC signal (< 1 μs)
`ARCLTDSTT` bits 3–6 or 7, 12–14 cleared	Cavity 1–4 window arc, air or vacuum side	AIM, on the Fast IC signal (< 1 μs)
`ARCLTDSTT` bit 8 cleared, `STN:MPS:LTCH` = FAULT	SPEAR MPS or orbit permit lost (External MPS)	VXI Slot 5 (`RF_FAULT`)
No arc latched, `HVPSXFORM:ARC:LTCH` set	HV transformer internal arc	SLC-500 HVPS PLC (~10–20 ms)
`HVPS:CROWBAR:LTCH` set, no arc latched	HVPS overvoltage → crowbar self-fired; OR analog crowbar threshold exceeded	Crowbar analog circuit (< 1 μs)
No arc latched, `STN:SUMY:PLC` = FAULT	RF MPS PLC trip (cooling water over-temp, vacuum excursion, or overpower)	RF MPS PLC (~10 ms; collector protection 0.70 s)
No arc latched, `HVPSOIL:TEMP:LTCH` set	Oil overtemperature (slow thermal fault)	SLC-500 HVPS PLC (~10–20 ms)
No arc latched, `HVPS:VOLT:LTCH` set	HVPS DC output overvoltage	SLC-500 HVPS PLC (~10–20 ms)
CH3 waveform entirely absent (oscilloscope)	SCR firing stopped — contactor open, or full SCR fault	Hardware, timing depends on cause
CH4 waveform asymmetric or one phase missing	Phase imbalance in primary; possible SCR failure or phase loss	SLC-500 or AC protection
CH1 voltage collapse: instant step to ~0 V	Crowbar fired	Crowbar circuit (trigger signal < 1 μs; the crowbar then **conducts** $\approx$ **10** μ**s** later, and the primary current is interrupted in **4–8 ms** — SLAC-PUB-7591)
CH1 voltage collapse: exponential decay over ~100 ms	SCR gate drive disabled (supervisory path)	SLC-500 relay (~10–20 ms)

Sources: §3.9 (AIM history buffer and SNL fault files); §7.5 (HVPS PV list); §10.6 (RF shutdown path); `rfApp/Db/rf_hvps.db` (HVPS scalar PV definitions); `llrf/llrf9/iGp/matlab/llrf/read_waveforms.m` (LabCA MATLAB access pattern); `iocBoot/b132-iocrf/Tables/device.tbl` (KSC2961 GPIB controller registration).

A EPICS PV Quick Reference

PV	Description	Normal State
{STN}:STNOFF:SUMY:STAT.SEVR	**Master trip wire** (SNL fault_stnoff)	NO_ALARM
{STN}:STN:AIM:MODU.HVPS	AIM HVPS permissive (aimon)	1 when ON
{STN}:STN:AIM:FRCBMABT	Force Beam Abort (fba)	0 when ON
{STN}:STN:AIM:MODU.RSTF	Reset AIM Faults (rstf)	— (write only)
{STN}:STN:AIM:MODU.RBA	Reset Beam Abort (rba)	— (write only)
{STN}:STN:AIM:ARCLTDSTT	Arc latched status, one bit per channel	0x71FB = nothing latched
{STN}:STN:AIM:ARCCURSTT	Arc current status, one bit per channel	0x71FB = no arc
{STN}:STN:AIM:ARCENBSTT	Arc fault enables (jumpers on the AIM)	0x71FB
{STN}:STN:AIM:FSTFLT	First-fault arc channel	0 = none
{STN}:STN:VXI:LTCH	VXI interlock latch (from AIM latch register)	OK
{STN}:STN:FORCED:LTCH	Forced fault latch (blocks auto-reset)	OK
{STN}:STN:MPS:LTCH	External MPS: the AIM's spare channel, from VXI Slot 5 (SPEAR MPS and orbit permits)	OK
{STN}:STN:SUMY:PLC	RF MPS PLC global interlock summary	OK
{STN}:HVPSSCR:ON:CTRL	HVPS SCR enable switch (hvpstrig)	1 when ON
{STN}:STN:RFP:RFENABLE	RF drive enable (rfswitch)	1 when ON
{STN}:HVPS:CROWBAR:LTCH	HVPS crowbar fired latch	OK
{STN}:HVPS:VOLT:LTCH	HVPS overvoltage latch	OK
{STN}:HVPSOIL:LEVEL:LTCH	HVPS oil level latch	OK
{STN}:HVPSOIL:TEMP:LTCH	HVPS oil temperature latch	OK
{STN}:HVPSXFORM:ARC:LTCH	Transformer arc latch	OK
{STN}:HVPS:PANIC:LTCH	Emergency Off latch	OK
{STN}:HVPSOFF:SUMY:STAT	HVPS OFF status summary	NO_ALARM
{STN}:HVPSSTN:SUMY:LTCH	HVPS station fault summary	NO_ALARM
{STN}:STNMPS:SUMY:LTCH	Station MPS summary (SPEAR3)	NO_ALARM

B Source Reference Index

All findings in this document were derived from direct analysis of the following source files. References are cited throughout as **[Rn]**.

B.1 Design Documents

Ref	File	Description
[R1]	`Designs/tex/docL-legacy-architecture/L_legacy_system_architecture.pdf`	SPEAR3 RF System Legacy Architecture (v2.5+) — primary system reference

B.2 Primary Source Code

Ref	File	Location	Lines	Role
[R2]	`rf_states.st,v`	`rfApp/src/seq/`	2,227	Master SNL state machine — all fault detection and response logic
[R3]	`devP2RfAim.c,v`	`rfApp/src/db/`	1,982	AIM module device support — arc channels, ISR, BATS, fault files
[R4]	`devABSLCDCM.c,v`	`allen Bradley/allen Bradley App/SLCDCM Src/`	563	SLC-500 Remote I/O device support — HVPS supervisory communication
[R5]	`p2RfCf2Def.h,v`	`rfApp/src/db/`	403	CF2 register definitions (PEP-II heritage, slot 5 reference)
[R6]	`p2RfAimDef.h,v`	`rfApp/src/db/`	—	AIM register definitions (`TRIPLVL`, `HISBUF`, `ARCVOL`, `FISTAT`, `HVPSON`)

B.3 EPICS Database Files

Ref	File	Location	Role
[R7]	`aim.db,v`	`rfApp/Db/`	AIM module PV definitions — arc status, interlock state, control outputs
[R8]	`rf_digital_hvps.db,v`	`rfApp/Db/`	HVPS digital I/O template — all HVPS status/latch bits from SLC-500
[R9]	`rf_digital_All.substitutions,v`	`rfApp/Db/`	Instantiates HVPS digital I/O with Remote I/O word/bit assignments
[R10]	`rf_sumy_hvps.db,v`	`rfApp/Db/`	HVPS alarm aggregation tree
[R11]	`rf_sumy_stn.db,v`	`rfApp/Db/`	Station-level alarm aggregation — STNOFF/STNON/STNMPS trees
[R12]	`rf_sumy_stn_spr.db,v`	`rfApp/Db/`	SPEAR3-specific additions to station summary (STNMPS, external MPS)
[R13]	`rf_interlock.db,v`	`rfApp/Db/`	DCM-based hardware interlock channels (MPS wiring)
[R14]	`rf_interlock_vxi.db,v`	`rfApp/Db/`	VXI-based interlock channels — AIM latch register bits
[R15]	`rf_analog_All.substitutions,v`	`rfApp/Db/`	Analog channel assignments (HVPS volts, current, temperature)
[R16]	`rf_hvps.db,v`	`rfApp/Db/`	HVPS supervisory records — voltage setpoints, SCR control, oil temp

Ref	File	Location	Role
[R17]	`crat_vxi_13slot.template,v`	`rfApp/Db/`	VXI crate template; the slot names are macros, filled by `srf1.substitutions,v` [R35]
[R18]	`rf_vxi_modules_All.substitutions,v`	`rfApp/Db/`	VXI module DB loading — `cf2.db` assigned to slot 5 (PEP-II artifact)

B.4 Technical Notes (Code Analysis Series)

Ref	File	Location	Content
[R19]	`00-executive-summary.md`	`codeReviewTechnicalNotes/`	System overview, slot 5 clarification
[R20]	`01-file-inventory.md`	`codeReviewTechnicalNotes/`	File classification — CF2 marked PEP-II only
[R21]	`03-vxi-device-support.md`	`codeReviewTechnicalNotes/`	AIM: arc channels, fast interlock chain, BATS, fault files
[R22]	`05-snl-state-machines.md`	`codeReviewTechnicalNotes/`	Complete 23-state SNL architecture documentation
[R23]	`06-plc-stepper-motors.md`	`codeReviewTechnicalNotes/`	PLC topology: 3 Remote I/O adapters, SLC-500 HVPS functions
[R24]	Switchgear and contactor drawings	`hvps/documentation/switchgear/`	GP-439-704-02-C1 (schematic), ID-308-801-06-C1 (connection wiring), GP-308-500-01-R3 (original LBL design, relay legend), Ross 713203 E-1 (HQ3 contactor and HCA-1-A driver). All verified by direct reading, September 2026
[R25]	`HoffmanBoxPPSWiring.docx`	`pps/`	J. Sebek — original terminal-by-terminal trace of the B118 Hoffman Box PPS wiring, switchgear theory of operation, and the K4/RR label-swap correction
[R26]	`LLRFDocumentationNotesR2.docx`	`llrf/documentation/`	J. Sebek — Remote I/O topology; the fibers between the AIM, the Fast Interlock Chassis, the RF Local Display Panel and the HVPS controller; the panel's cross-connects
[R27]	`fiberOpticCableSignalControlRev3.docx`	`llrf/documentation/`	J. Sebek — the three fibers between the LLRF system and the HVPS controller, their light states and the inputs of the panel logic
[R28]	`controllerFiberOpticConnections.docx`	`hvps/architecture/designNotes/`	J. Sebek — HVPS controller side: slot-5 output 0 (24 V fault/enable) on the COMMANDS bus of the trigger interconnect boards
[R29]	RF Local Display Panel drawings	`llrf/documentation/localPanel/`	DL-340-310-00, front and rear panels SI-340-310-05 and -02, schematic SD-340-311-01 (GAL22V10 logic, fiber transceivers, beam-abort BNC J6)
[R30]	Wiring diagrams WD-340-330-29 and -02	`llrf/documentation/mpsWiringDiagrams/`	Control fiber-optic cables of the RF Local Display Panel; cross-connect X530 to its connectors J2 and J3
[R31]	`SP3_writeup_V1d2.pdf`	`llrf/documentation/legacyArchitecture/`	J. Dusatko, SPEAR3 LLRF system description, v1.2 — the AIM's six control signals, set by register bits and its on-board fault logic; the MPS interlock module, which drives RF_FAULT and has no VXIbus interface

Ref	File	Location	Content
[R32]	Fast Interlock Chassis and AIM schematics	`llrf/documentation/legacyInterfaceModules/`	SD-340-308-01 and -02 (Fast IC: photodetectors and peak detectors, command fibers, RF detector; J2, J15, J16, J31, J32) and SD-340-309-01 (AIM: comparators with DAC thresholds, latches, fast ADC, fault-enable jumper headers H5 and H6; J3, J4)
[R33]	PEP-II RF station block diagram BD-340-330-00	`llrf/documentation/legacyArchitecture/`	P. Corredoura, 1999: the local panel's beam-abort output leaves the station for the PEP-II beam-abort system
[R34]	`llrfInterfaceChassis.docx`	`llrf/architecture/`	J. Sebek, LLRF Interface Chassis specification — the SPEAR MPS and the orbit interlock each give the LLRF a 24 V permit
[R35]	SPEAR3 station database: `rf_stn_All.substitutions,v`, `rf_stn_4CVAll.substitutions,v`, `rf_interlock_arc.db,v`; `srf1.substitutions,v`	`rfApp/Db/`; `rfApp/DbIoc/`	The AIM arc channels and their names, with External MPS on the spare channel since 2003; the slot names of the SRF1 crate (slot 5 MPS Shutoff, slot 6 Link Passthru)
[R36]	Wiring diagrams WD-340-330-23 and -25	`llrf/documentation/mpsWiringDiagrams/`	RF MPS PLC-5 digital inputs and outputs
[R37]	RF MPS PLC-5 program (2011–2012), decoded	`llrf/allenBradley/decoded/`	Collector protection: HVPS voltage times current minus the square of 0.017 times the RF detector reading, above 1 200 kW for 0.70 s, latches and drops the collector output and the global interlock summary